When carry crowds: endogenous foreign exchange crash risk, capital inflows, and foreign-investor coordination

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Abstract

This paper develops a tractable model of crowded carry trades in which foreign capital inflows endogenously increase exchange-rate crash risk. A continuum of atomistic foreign investors chooses local-currency positions while taking aggregate inflows as given, although aggregate positioning raises both expected currency appreciation over the relevant holding period and the probability of a discrete depreciation. The symmetric competitive equilibrium is characterized by a cubic fixed-point equation with closed-form solutions. With positive baseline carry, the equilibrium is unique. When baseline carry is weak and the appreciation feedback is sufficiently strong, however, a no-inflow equilibrium can coexist with a self-sustaining inflow boom, so a deterioration in global funding or risk conditions can trigger a discontinuous sudden stop. Importantly, the additional high-inflow equilibrium is generated by the positive appreciation feedback, while endogenous crash risk limits the boom rather than generating multiplicity by itself. The competitive allocation also differs from a foreign-investor-sector coordination benchmark because atomistic investors fail to internalize the systemic fragility created by their positions. In high-inflow states, fragility dominates and the market takes excessive tail risk relative to this benchmark. A prudential wedge can reduce inflows and crash probability and can eliminate fragile multiplicity. The framework provides a simple analytical link between crowded carry trades, endogenous tail risk, global funding conditions, and macroprudential policy.

Keywords: Carry trades; Endogenous crash risk; Capital inflows; Sudden stops; Macroprudential policy

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1 Introduction

Currency carry trades are attractive precisely because they often look uneventful. An investor borrows in a low-interest-rate funding currency, buys a higher-yielding local-currency asset, and earns the interest differential so long as the investment currency does not depreciate too much. The empirical difficulty is that the quiet periods are punctuated by rare, abrupt losses. High-interest-rate currencies tend to load on crash risk and negative skewness, so average carry returns can partly compensate investors for exposure to infrequent but severe depreciations [Brunnermeier et al., 2009, Jurek, 2014, Farhi et al., 2009]. The peso-problem interpretation is therefore central: an apparently favorable mean return may coexist with a low-probability state that is difficult to infer from short samples [Burnside et al., 2011].

This paper asks what changes when the probability of that crash is not fixed, but rises with the aggregate scale of the trade itself. The mechanism is deliberately simple. A continuum of atomistic foreign investors chooses positions in an emerging-market (EM) local-currency security. Each investor is too small to affect the aggregate state and therefore takes the exchange-rate distribution as given. In the aggregate, however, crowded positioning increases the probability of a discrete depreciation. Large inflows can be supported by favorable carry and, when present, persistent portfolio-balance pressure on the currency; the same positions simultaneously make the market more fragile.

This distinction between individual choice and aggregate positioning is the organizing principle of the model. Investor i chooses f_i, while aggregate foreign positioning is $F = \int_0^1 f_i \, di$. The private problem is strictly concave and has a unique best response. Equilibrium consistency, $f_i = F$ in a symmetric allocation, introduces the nonlinearity: the competitive equilibrium is characterized by a cubic fixed-point equation. This separation makes it possible to study existence, uniqueness, multiplicity, tail risk, and policy analytically without assigning a single investor control over a market-wide fragility state.

The first result is a sharp separation between fragility and multiplicity. With positive baseline carry, the competitive equilibrium is unique. If baseline carry is negative at zero inflows, an additional high-inflow equilibrium can arise only when aggregate positioning raises expected holding-period appreciation strongly enough to overcome balance-sheet and baseline-risk costs. Endogenous crash risk does not create the additional equilibrium; instead, the convex crash-risk terms eventually bend the best-response map back and cap the boom.[1] The multiplicity result is therefore

[1]The paper makes this scope condition explicit, derives the minimum appreciation coefficient required for the feedback to become locally self-reinforcing, and reports a simple sensitivity exercise.

presented as a disciplined strong-feedback case, while the unique-equilibrium crowding and fragility results apply much more broadly.

The appreciation term requires careful timing. Standard portfolio-balance models can associate an inflow with an immediate appreciation that is already reflected in the entry exchange rate, followed by expected depreciation [Gabaix and Maggiori, 2015]. The coefficient χ in this paper is not intended to relabel that contemporaneous price impact. It captures any additional expected exchange-rate movement over the holding period after the entry rate is set: for example, gradual implementation of fund reallocations, persistent customer order flow, slow dealer inventory absorption, or other delayed portfolio-balance pressure. Accordingly, the model allows $\chi \in \mathbb{R}$. Positive χ is the special case needed for self-sustaining inflow multiplicity; $\chi = 0$ eliminates the appreciation feedback; and $\chi < 0$ captures expected reversal or depreciation after entry. Empirically, exogenous portfolio reallocations have been shown to move currencies [Hau et al., 2010], weekly portfolio inflows to EMs are associated with exchange-rate appreciation [Ananchotikul and Zhang, 2014], and EM bond inflows predict subsequent currency appreciation [Hördahl and Valente, 2022]. These findings motivate treating the sign and persistence of the holding-period flow effect as an empirical object rather than imposing it mechanically.

The second result concerns coordination among the foreign investors. Because aggregate positioning changes the return distribution faced by every investor, the decentralized allocation generally differs from the position that would be chosen by a coordinator maximizing the aggregate mean–variance payoff of the foreign-investor sector. This is intentionally not a host-country social-welfare criterion. An appreciation raises foreign investors' returns, but the counterpart of that price movement lies elsewhere in the economy; without modeling domestic agents or another distortion, that price effect cannot be counted as an unqualified social benefit. The paper therefore calls the appreciation component a foreign-sector price spillover, not a welfare-relevant pecuniary externality. By contrast, the endogenous crash-probability and variance terms are genuine within-sector fragility spillovers: each atomistic investor ignores how a larger position raises the tail risk borne by all foreign investors. Under an explicit uniqueness or concavity condition, the sign of the resulting coordination gap ranks the competitive position relative to the foreign-sector benchmark.

The third result concerns policy, but with equally narrow interpretation. A wedge on foreigners' after-cost return can reduce equilibrium inflows and crash probability when the direct reduction in expected return dominates the mechanical reduction in measured variance; the paper states the exact parameter restriction required for this comparative static. Under additional continuity and crossing conditions, such a wedge can decentralize the foreign-investor coordination benchmark. This is an im-

plementation result, not a claim that the wedge is optimal for the host country. If the wedge is a tax, its revenue accrues to the host-country public sector unless otherwise specified; a lump-sum rebate to foreign investors leaves their marginal condition unchanged, whereas a position-proportional rebate would undo the wedge. A host-country welfare analysis would require a domestic side of the model.

The closest empirical and theoretical motivation comes from the carry-trade, crash-risk, and peso-problem literatures [Brunnermeier et al., 2009, Burnside et al., 2011, Jurek, 2014, Farhi et al., 2009, Menkhoff et al., 2012]. The model is also related to portfolio-balance and intermediary theories in which limited risk-bearing makes exchange rates respond to financial demand [Gabaix and Maggiori, 2015, Hau et al., 2010]. A separate connection is to work on constrained-inefficient credit booms and systemic externalities [Lorenzoni, 2008, Bianchi, 2011, Mendoza, 2010, Jeanne and Korinek, 2010, Korinek, 2011, Bianchi and Mendoza, 2018]. The analogy is useful for the fragility channel, but the normative object here is deliberately narrower: the coordinator represents foreign investors, not a benevolent government of the recipient economy.

The global financial cycle plays a background role. The scalar state g shifts foreign funding costs, risk tolerance, baseline volatility, and baseline crash risk, consistent with evidence on global funding and risk conditions [Rey, 2015, Bruno and Shin, 2015, Miranda-Agrippino and Rey, 2020]. The model does not attempt to explain cross-country comovement in asset prices or to generate the global financial cycle endogenously. It is a one-country model of how an exogenous common funding/risk environment maps into foreign positioning and endogenous FX tail risk.

The paper makes three contributions. First, it provides a closed-form decentralized model in which crash risk is endogenous to aggregate carry positioning while each investor has a well-behaved private problem. Second, it identifies precisely what can and cannot generate multiplicity: positive delayed appreciation feedback creates the additional high-inflow equilibrium, while endogenous fragility limits it; when the feedback is zero or negative, the multiplicity disappears. Third, it derives a transparent foreign-investor coordination gap, separates the price spillover from genuine fragility spillovers, and states the conditions under which a prudential wedge lowers inflows, lowers crash risk, and can implement the coordination benchmark.

Section 2 presents the environment, timing, private best responses, equilibrium, the foreign-investor coordination benchmark, multiplicity, and comparative statics. Section 3 concludes. Proofs are collected in Appendix A; Appendix B develops a margin-style instrument.

2 Model

2.1 Set up

Time is $t \in \{0, 1\}$. There is a unit continuum of identical, atomistic foreign investors indexed by $i \in [0, 1]$, and a single emerging-market (EM) local-currency security. The model contains no domestic household, firm, or government objective; all private payoffs below are foreign-investor payoffs. Throughout, variables are expressed in gross form.

Let $e_t > 0$ denote the spot exchange rate, defined as foreign currency per unit of EM currency, so an increase in e is an EM-currency appreciation from a foreign investor's perspective. A scalar state $g \in (0, 1]$ summarizes exogenous global funding and risk conditions; higher g means easier conditions. It affects the foreign funding gross rate $R^*(g) > 1$, the investors' mean–variance risk-aversion coefficient $\lambda(g) > 0$, normal-times FX volatility $\sigma_\varepsilon(g) > 0$, and the baseline crash probability $\bar{\pi}(g) \in (0, 1)$. Section 2.11 imposes sign restrictions. The state g is not an endogenous global-financial-cycle factor generated by this one-country model; it is a reduced-form common shifter taken as given.

At $t = 0$ investor i chooses a nonnegative foreign-currency notional $f_i \geq 0$ in a local-currency security with gross local return $R > 1$ between $t = 0$ and $t = 1$. Aggregate foreign positioning is

$$F \equiv \int_0^1 f_i \, di.$$

The scale of F is a normalization. For quantitative interpretation one may, for example, measure one unit of F as portfolio flow equal to 0.1% of host-country GDP; changing that unit simply rescales χ, η, and κ and leaves the analytical results unchanged.

Investor i funds the position at $R^*(g)$. A policy wedge $\tau \in [0, 1)$ reduces the effective local return received by foreigners. The wedge can represent a tax on foreign holdings, an unremunerated reserve requirement, or a regulatory cost. Its incidence is discussed explicitly in Section 2.6. The investor's foreign-currency net excess payoff is

$$K_i \equiv f_i \left[(1 - \tau) \frac{e_1}{e_0} R - R^*(g) \right] - \frac{\kappa}{2} f_i^2, \tag{1}$$

where $\kappa > 0$ is a convex balance-sheet, intermediation, or internal position cost.

2.2 The exchange rate, timing, and aggregate fragility

The entry exchange rate e_0 is the spot rate at which the holding period begins. The aggregate state F is the stock of foreign positioning associated with that investment episode. Equation (2) allows the same aggregate positioning to forecast an additional exchange-rate movement between the entry rate e_0 and the liquidation rate e_1:

$$\frac{e_1}{e_0} = \underbrace{1 + \mu_e(g) + \chi F + \varepsilon}_{\text{non-crash holding-period component}} \underbrace{- J\mathbf{1}\{\mathcal{C} = 1\}}_{\text{crash jump}}. \tag{2}$$

Here $\mu_e(g) \in \mathbb{R}$ is baseline expected appreciation, ε has $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma_\varepsilon^2(g)$, and $J > 0$ is the size of a discrete depreciation in the crash state. Because e is foreign currency per unit of EM currency, the crash enters with a minus sign.

The parameter $\chi \in \mathbb{R}$ is a holding-period flow-feedback coefficient. It should not be interpreted as the contemporaneous price impact that has already been capitalized into e_0. Standard portfolio-balance logic can produce immediate appreciation on entry followed by expected depreciation; that case is consistent here with $\chi \leq 0$. The case $\chi > 0$ instead represents persistent or delayed demand pressure after the entry rate is set—for example, gradual fund implementation, persistent customer order flow, slow intermediary inventory absorption, or continuing portfolio rebalancing. Evidence that exogenous portfolio reallocations move exchange rates and that EM inflows are associated with both contemporaneous and subsequent appreciation provides motivation for considering this case [Hau et al., 2010, Ananchotikul and Zhang, 2014, Hördahl and Valente, 2022]. The multiplicity result below will require $\chi > 0$; all baseline uniqueness and endogenous-fragility results do not.

The crash probability depends on aggregate positioning:

$$\mathbb{P}(\mathcal{C} = 1 \mid F, g) = \pi(F, g) \equiv \bar{\pi}(g) + \eta F^2, \tag{3}$$

with $\eta > 0$. The quadratic form captures the idea that crowded positions, leverage, maturity mismatch, or dealer balance-sheet constraints make large inflow states disproportionately fragile while preserving a tractable polynomial equilibrium.

Remark 1 (Individual choice versus aggregate state)**.** Both χF in Equation (2) and $\pi(F, g)$ in Equation (3) depend on aggregate foreign positioning, not on any individual f_i. Each investor is of measure zero and treats these objects as given when choosing f_i. Symmetric equilibrium later imposes the consistency condition $f_i = F$.

Assumption 1 (Feasible crash probability)**.** There exists $\bar{F} > 0$ such that the relevant equilibrium and benchmark allocations lie in $[0, \bar{F}]$ and $\pi(F, g) \in (0, 1)$ on this set.

Since $\pi(F,g) = \bar{\pi}(g) + \eta F^2$, a sufficient upper bound satisfies $\eta \bar{F}^2 < 1 - \sup_g \bar{\pi}(g)$.

2.3 The individual problem and the best response

Investor i is mean–variance. Let $\theta > 0$ be a taste shifter on expected payoff (a "mean weight") and $\lambda(g) > 0$ risk aversion. Investor i chooses $f_i \geq 0$ to maximize

$$U_i(f_i; F, g, \tau) \;\equiv\; \theta\, \mathbb{E}[K_i \mid f_i, F, g] \;-\; \frac{\lambda(g)}{2} \operatorname{Var}(K_i \mid f_i, F, g)\,, \tag{4}$$

taking the aggregate F as given. Higher θ increases desired exposure; higher $\lambda(g)$ reduces it, and λ moves with global conditions.

Define the random per-unit excess return in foreign currency, common to all investors and a function of the aggregate state,

$$X(F, g, \tau) \;\equiv\; (1-\tau)\,\frac{e_1}{e_0}\,R - R^*(g). \tag{5}$$

Then $K_i = f_i X - \frac{\kappa}{2} f_i^2$, and conditional on (F, g, τ),

$$\mathbb{E}[K_i] = f_i\,\mu_X(F,g,\tau) - \frac{\kappa}{2} f_i^2, \qquad \operatorname{Var}(K_i) = f_i^2\,\sigma_X^2(F,g,\tau), \tag{6}$$

where $\mu_X = \mathbb{E}[X]$ and $\sigma_X^2 = \operatorname{Var}(X)$ depend only on the aggregate F. Using Equation (2)–(3) and the independence of ε and $\mathcal{C}$,

$$\mu_X(F,g,\tau) = (1-\tau)R\big(1 + \mu_e(g) + \chi F - J\pi(F,g)\big) - R^*(g), \tag{7}$$

$$\sigma_X^2(F,g,\tau) = (1-\tau)^2 R^2 \Big(\sigma_\varepsilon^2(g) + J^2 \pi(F,g)\big(1 - \pi(F,g)\big)\Big). \tag{8}$$

The term $J\pi(F,g)$ in Equation (7) is a crash-expected-loss adjustment; $J^2\pi(1-\pi)$ in Equation (8) is the jump contribution to variance. For closed-form characterization we use the standard rare-crash approximation.

Assumption 2 (Rare crash)**.** In the relevant equilibrium region, $\pi(F,g)$ is small enough that $1 - \pi(F,g) \approx 1$ in the second-order term of Equation (8), so that

$$\sigma_X^2(F,g,\tau) \;\approx\; (1-\tau)^2 R^2 \big(\sigma_\varepsilon^2(g) + J^2 \pi(F,g)\big). \tag{9}$$

Remark 2. Assumption 2 preserves the qualitative economics while yielding a polynomial equilibrium condition. Dropping it and keeping $\pi(1-\pi)$ leaves the proofs intact at the cost of heavier algebra (the equilibrium polynomial rises in degree).

It is convenient to collect the following composite coefficients:

$$A(g,\tau) \equiv (1-\tau)R\big(1+\mu_e(g)-J\bar{\pi}(g)\big)-R^*(g), \tag{10}$$

$$B(\tau) \equiv (1-\tau)R\,\chi, \tag{11}$$

$$C(g,\tau) \equiv (1-\tau)R\,J\eta, \tag{12}$$

$$S(g,\tau) \equiv (1-\tau)^2R^2\big(\sigma_\varepsilon^2(g)+J^2\bar{\pi}(g)\big), \tag{13}$$

$$D(\tau) \equiv (1-\tau)^2R^2J^2\eta. \tag{14}$$

Here A is baseline expected excess carry (net of baseline crash losses); B is the strength of inflow-driven expected appreciation; C is the marginal expected crash loss induced by inflows; S is baseline variance; and D is how strongly inflows raise variance via endogenous crash probability. With these, the conditional mean and variance of X are

$$\mu_X(F,g,\tau) = A+BF-CF^2, \qquad \sigma_X^2(F,g,\tau) = S+DF^2. \tag{15}$$

Investor i's objective becomes

$$U_i(f_i;F,g,\tau) = \theta\Big(f_i\,\mu_X(F,g,\tau)-\frac{\kappa}{2}f_i^2\Big)-\frac{\lambda(g)}{2}\,f_i^2\,\sigma_X^2(F,g,\tau). \tag{16}$$

Because μ_X and σ_X^2 are taken as given, U_i is strictly concave quadratic in f_i, with a unique best response.

Lemma 1 (Linear best response)**.** *For any (F,g,τ), investor i's problem (16) is strictly concave in f_i, and the unique best response is*

$$\mathcal{B}(F;g,\tau) \;=\; \max\left\{0,\;\frac{\theta\,\mu_X(F,g,\tau)}{\theta\kappa+\lambda(g)\,\sigma_X^2(F,g,\tau)}\right\}. \tag{17}$$

Proof. See Appendix A. □

The economics is transparent. The atomistic investor equates the marginal expected gain $\theta\mu_X$ to the marginal cost from the balance-sheet penalty $\theta\kappa f_i$ plus the marginal risk penalty $\lambda\sigma_X^2 f_i$. Crash risk and appreciation enter only through μ_X and σ_X^2, which the investor cannot move. The private first-order condition contains no derivative of the aggregate state: appreciation and crash risk affect the investor through the perceived mean and variance, while their dependence on aggregate positioning reappears only through equilibrium consistency.

2.4 Competitive equilibrium: the cubic that governs everything

We use the symmetric competitive equilibrium throughout. It is the natural decentralization of the carry trade: identical small investors best respond to an aggregate they do not control, and consistency requires that the common choice equal the aggregate.

Definition 1 (Symmetric competitive equilibrium)**.** A symmetric competitive equilibrium (CE) is an inflow level $F^{\mathrm{CE}}(g,\tau) \geq 0$ such that every investor chooses $f_i = F^{\mathrm{CE}}$ optimally given the aggregate $F = F^{\mathrm{CE}}$; that is, $F^{\mathrm{CE}} = \mathcal{B}(F^{\mathrm{CE}}; g, \tau)$.

Imposing $f_i = F$ on the best response (17) (interior case $\mu_X > 0$) gives $F\big(\theta\kappa + \lambda\sigma_X^2(F)\big) = \theta\mu_X(F)$. Substituting (15) and rearranging yields the equilibrium cubic:

$$\begin{aligned}\lambda(g)D(\tau)F^3 + \theta C(g,\tau)F^2 \\ + \big[\theta\kappa + \lambda(g)S(g,\tau) - \theta B(\tau)\big]F - \theta A(g,\tau) = 0.\end{aligned} \tag{18}$$

We write Equation (18) compactly as $H(F; g, \tau) = 0$ with

$$\begin{aligned}H(F;g,\tau) &\equiv a_3F^3 + a_2F^2 + a_1F + a_0, \\ a_3 &= \lambda(g)D(\tau) > 0, \qquad a_2 = \theta C(g,\tau) > 0, \\ a_1 &= \theta\kappa + \lambda(g)S(g,\tau) - \theta B(\tau), \qquad a_0 = -\theta A(g,\tau).\end{aligned} \tag{19}$$

The following subsection records existence, uniqueness, and the closed form for the competitive cubic (18). Aggregate effects that private investors do not internalize are isolated later through the foreign-investor coordination benchmark in Section 2.6.

2.5 Existence, uniqueness, and closed form

Lemma 2 (Existence of an interior competitive equilibrium)**.** *Suppose $a_3 > 0$ and $A(g,\tau) > 0$. Then*

$$H(0;g,\tau) < 0, \qquad \lim_{F\to\infty} H(F;g,\tau) = +\infty.$$

Hence Equation (18) has at least one strictly positive root and an interior competitive equilibrium exists.

Proof. See Appendix A. □

The condition $A > 0$ says that baseline carry is attractive even at zero positioning, so the market does not collapse to no inflows. The economics of the limit is straightforward: as aggregate inflows grow without bound, equilibrium variance scales with

F^2 (through DF^2) and crash-loss drag scales with F^2 (through CF^2), so the marginal investor's perceived return turns negative and inflows cannot be sustained.

Proposition 1 (Uniqueness with positive baseline carry)**.** *If $A(g,\tau) > 0$, the competitive equilibrium is unique. More generally, a sufficient condition for at most one positive root of (18) is $a_2 > 0$ and $a_0 \le 0$, which holds whenever $C > 0$ and $A \ge 0$.*

Proof. See Appendix A. □

The proof (Appendix A) is a one-line application of Descartes' rule of signs to (19): with $a_3 > 0$, $a_2 > 0$, $a_0 < 0$, and a_1 of either sign, the coefficient sequence exhibits exactly one sign change, so there is exactly one positive root. This delivers a clean and economically important message that the convexity of crash risk in inflows steepens the cost of large positions but does not, on its own, create a second equilibrium. Multiplicity requires a different configuration, analyzed in Section 2.7.

Closed-form roots. Write Equation (18) in the form $a_3F^3 + a_2F^2 + a_1F + a_0 = 0$ and apply the depressed-cubic (Tschirnhaus) transformation $F = y - a_2/(3a_3)$ to obtain

$$y^3 + p\,y + q = 0, \qquad p = \frac{3a_3a_1 - a_2^2}{3a_3^2}, \qquad q = \frac{2a_2^3 - 9a_3a_2a_1 + 27a_3^2a_0}{27a_3^3}, \tag{20}$$

with discriminant $\Delta = (q/2)^2 + (p/3)^3$.

Proposition 2 (Characterization of the real roots)**.** *The depressed cubic (20) has exactly one real root if $\Delta > 0$, three distinct real roots if $\Delta < 0$, and a repeated real root if $\Delta = 0$. Any real root y maps to a real F via $F = y - a_2/(3a_3)$. When $\Delta > 0$ the unique real root is*

$$y^\star = \sqrt[3]{-\tfrac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\tfrac{q}{2} - \sqrt{\Delta}}, \qquad F^{\mathrm{CE}} = y^\star - \frac{a_2}{3a_3}, \tag{21}$$

and when $\Delta < 0$ the three real roots admit the trigonometric representation

$$y_k = 2\sqrt{-\tfrac{p}{3}}\,\cos\!\left(\tfrac{1}{3}\arccos\!\left(\tfrac{3q}{2p}\sqrt{-\tfrac{3}{p}}\right) - \tfrac{2\pi k}{3}\right), \quad k = 0,1,2, \tag{22}$$

with $F_k = y_k - a_2/(3a_3)$. The economically relevant equilibria are the nonnegative roots; in the multiplicity case (Section 2.7) stability selects among them.

Proof. See Appendix A. □

2.6 Foreign-investor coordination benchmark

The decentralized equilibrium can be compared with a benchmark in which a coordinator chooses the common position of the foreign-investor sector and internalizes how aggregate positioning changes the return distribution faced by that sector. This benchmark is deliberately narrower than social welfare. There is no domestic side in the model, so the coordinator does not represent the host-country government and W^{FI} below should not be read as host-country welfare.

For the benchmark, set the policy wedge to zero and define $A_0 \equiv A(g,0)$, $B_0 \equiv B(0)$, $C_0 \equiv C(g,0)$, $S_0 \equiv S(g,0)$, and $D_0 \equiv D(0)$. The foreign-investor coordinator chooses F to maximize the aggregate mean–variance criterion

$$W^{\mathrm{FI}}(F;g) = \theta\left[F\mu_X(F,g,0) - \frac{\kappa}{2}F^2\right] - \frac{\lambda(g)}{2}F^2\sigma_X^2(F,g,0), \tag{23}$$

which is

$$W^{\mathrm{FI}}(F;g) = \theta\left(A_0F + B_0F^2 - C_0F^3 - \frac{\kappa}{2}F^2\right) - \frac{\lambda(g)}{2}F^2(S_0 + D_0F^2). \tag{24}$$

Its first-order condition is

$$W_F^{\mathrm{FI}}(F;g) = \theta\big[A_0 + (2B_0 - \kappa)F - 3C_0F^2\big] - \lambda(g)\big(S_0F + 2D_0F^3\big) = 0. \tag{25}$$

Let $F^{\mathrm{FI}}(g)$ denote a global maximizer on the feasible set.

To compare private and coordinated incentives, evaluate the private marginal value at $\tau = 0$ in a symmetric configuration:

$$\mathrm{MV}_0^{\mathrm{priv}}(F) \equiv \theta\mu_X(F,g,0) - \theta\kappa F - \lambda(g)F\sigma_X^2(F,g,0).$$

The unregulated competitive equilibrium satisfies $\mathrm{MV}_0^{\mathrm{priv}}(F^{\mathrm{CE}}(g,0)) = 0$ when interior.

Proposition 3 (Foreign-investor coordination gap)**.** *The difference between the coordinator's marginal objective and the private marginal value is*

$$\delta^{\mathrm{FI}}(F) \equiv W_F^{\mathrm{FI}}(F) - \mathrm{MV}_0^{\mathrm{priv}}(F) = \theta B_0F - 2\theta C_0F^2 - \lambda(g)D_0F^3. \tag{26}$$

Proof. See Appendix A. □

The three terms have different interpretations. The component $+\theta B_0F$ is a foreign-sector price spillover: when $\chi > 0$, aggregate positioning raises the holding-period

appreciation received by other foreign investors. It is a group return effect, but it is not by itself a host-country social benefit; an exchange-rate appreciation has a counterpart for domestic agents or other market participants that the model does not include. The terms $-2\theta C_0 F^2$ and $-\lambda(g) D_0 F^3$ are genuine within-sector fragility spillovers. Each investor ignores how an additional position raises, respectively, the expected crash loss and the jump-risk variance borne by all foreign investors.

Proposition 4 (Ranking relative to the foreign-investor benchmark)**.** *Suppose the unregulated competitive equilibrium is interior. Assume either (i) $A_0 > 0$, or (ii) W^{FI} is strictly concave on the feasible set. Then the coordinator has a unique relevant maximizer F^{FI}, and the sign of $\delta^{\mathrm{FI}}(F^{\mathrm{CE}})$ ranks the two allocations. If $B_0 > 0$, define*

$$\widehat{F} \equiv \frac{-\theta C_0 + \sqrt{\theta^2 C_0^2 + \lambda(g) D_0 \theta B_0}}{\lambda(g) D_0}. \tag{27}$$

Then $F^{\mathrm{CE}} > \widehat{F}$ implies $F^{\mathrm{FI}} < F^{\mathrm{CE}}$, while $F^{\mathrm{CE}} < \widehat{F}$ implies $F^{\mathrm{FI}} > F^{\mathrm{CE}}$. If $B_0 \leq 0$, then $\delta^{\mathrm{FI}}(F) < 0$ for every $F > 0$, so $F^{\mathrm{FI}} < F^{\mathrm{CE}}$.

Proof. See Appendix A. □

The qualifier in Proposition 4 matters. A negative derivative of a non-concave objective at the competitive allocation is not, by itself, enough to rank a global optimum. Condition (i) is particularly useful here: $W_F^{\mathrm{FI}}(0) = \theta A_0 > 0$, $W_F^{\mathrm{FI}}(F) \to -\infty$, and W_F^{FI} is strictly concave because $W_{FFF}^{\mathrm{FI}} = -6\theta C_0 - 12\lambda(g) D_0 F < 0$. Hence its positive zero is unique. Condition (ii) provides the standard alternative.

Proposition 5 (Implementation by a prudential wedge)**.** *Suppose $F^{\mathrm{CE}}(g, 0) > F^{\mathrm{FI}}(g) > 0$. Suppose further that along the relevant unique equilibrium branch the conditions of Proposition 7 hold, so $F^{\mathrm{CE}}(g, \tau)$ is continuous and strictly decreasing in τ, and that there exists $\bar{\tau} \in (0, 1)$ with $F^{\mathrm{CE}}(g, \bar{\tau}) < F^{\mathrm{FI}}(g)$. Then there is a unique $\tau^\star \in (0, \bar{\tau})$ such that $F^{\mathrm{CE}}(g, \tau^\star) = F^{\mathrm{FI}}(g)$.*

Proof. See Appendix A. □

This proposition is an implementation statement, not a host-country optimal-tax result. If τ is a literal tax, revenue is paid by foreign investors and accrues to the host public sector; that transfer is outside W^{FI}. If revenue is instead rebated lump-sum to the foreign-investor sector in a way that is independent of an individual investor's own f_i, the private first-order condition is unchanged, although the level of sector resources changes. A position-proportional rebate would undo the marginal wedge. A full welfare analysis would need to model domestic incidence, the use of revenue, and the domestic consequences of appreciation and crashes.

2.7 Multiple equilibria and self-fulfilling sudden stops

Multiplicity is not generated by endogenous crash risk alone. Proposition 1 already implies that when baseline carry is positive, the competitive equilibrium is unique. When baseline carry is negative, a second, high-inflow equilibrium requires the holding-period appreciation channel to create a sufficiently strong positive feedback. The fragility terms then limit the boom at large F.

For $a_1 < 0$, the positive turning point of the equilibrium polynomial is

$$F_m \equiv \frac{-a_2 + \sqrt{a_2^2 - 3a_3a_1}}{3a_3} > 0. \tag{28}$$

Theorem 1 (Appreciation-feedback multiplicity). *Suppose $A(g,\tau) < 0$ and*

$$\theta B(\tau) > \theta\kappa + \lambda(g)S(g,\tau), \tag{29}$$

so $a_1 < 0$. If, in addition, $H(F_m; g, \tau) < 0$, then the model has exactly three nonnegative competitive equilibria: $F = 0$ and two interior roots $0 < F_L < F_m < F_H$. Under continuous best-response adjustment $\dot{F} = \mathcal{B}(F; g, \tau) - F$, $F = 0$ and F_H are locally stable and F_L is unstable. A parameter change that raises the local minimum to zero removes F_L and F_H in a saddle-node bifurcation, generating a discontinuous switch from the boom equilibrium to $F = 0$.

Proof. See Appendix A. □

Condition (29) has a transparent interpretation. Since $B(\tau) = (1-\tau)R\chi$, it is equivalent to

$$\chi > \chi_{\min}(g,\tau) \equiv \frac{\kappa + [\lambda(g)/\theta]S(g,\tau)}{(1-\tau)R}. \tag{30}$$

The expected holding-period appreciation generated by one unit of aggregate positioning must be large enough to offset the marginal balance-sheet cost κ and the baseline variance penalty $[\lambda/\theta]S$. This inequality is necessary for the cubic to slope downward at the origin when $A < 0$; the additional condition $H(F_m) < 0$ ensures that the downward movement is large enough to create the two interior crossings. Thus, Equation (29) should not be read as a stand-alone sufficient condition for multiplicity.

Illustrative magnitudes. To give Equation (30) scale, normalize one unit of F to a portfolio-flow shock equal to 0.1% of recipient-country GDP and set $\theta = 1$, $R = 1.02$, $\tau = 0$, $\sigma_\varepsilon = 0.02$, $J = 0.10$, and $\bar{\pi} = 0.01$. These values imply $S = 0.0005202$. Table 1

reports the minimum χ required for $a_1 < 0$ across risk aversion and balance-sheet costs. The entries are percentage appreciation per unit of normalized flow.

Table 1: Minimum delayed-appreciation feedback required for $a_1 < 0$

	$\lambda = 2$	$\lambda = 4$	$\lambda = 6$
$\kappa = 0.002$	0.30%	0.40%	0.50%
$\kappa = 0.005$	0.59%	0.69%	0.80%
$\kappa = 0.010$	1.08%	1.18%	1.29%

Notes: The table evaluates Equation (30) under the normalization and parameters in the text. It is a sensitivity calculation, not a calibration. The condition $a_1 < 0$ is necessary but not sufficient for multiplicity; $A < 0$ and $H(F_m) < 0$ are also required.

The scale is economically informative. Ananchotikul and Zhang [2014] report that, in normal times, a weekly portfolio-flow increase equal to 0.1% of GDP is associated on average with roughly 0.4–0.8% exchange-rate appreciation across EM regions, with substantially larger effects during crisis periods. Hau et al. [2010] identify currency appreciation around exogenous MSCI-induced equity reallocations, and Hördahl and Valente [2022] find that bond inflows predict subsequent EME currency appreciation. These studies do not identify the structural χ in this model, and contemporaneous price impact need not equal delayed appreciation. They nevertheless show that the order of magnitude needed for the local feedback in some rows of Table 1 is not mechanically implausible. At the same time, the higher-κ rows are demanding, and the additional conditions $A < 0$ and $H(F_m) < 0$ make multiplicity a selective region rather than the model's generic prediction.

Remark 3 (What survives when $\chi \leq 0$)**.** If $\chi = 0$, then $B = 0$; if $\chi < 0$, then $B < 0$. In either case $a_1 = \theta\kappa + \lambda S - \theta B > 0$, so $H'(F) > 0$ for every $F > 0$. Hence the competitive equilibrium is unique and positive when $A > 0$, and the only equilibrium is $F = 0$ when $A \leq 0$. Endogenous crash risk, the risk-price results, and the foreign-investor coordination analysis remain. Moreover, $B_0 \leq 0$ implies $\delta^{\mathrm{FI}}(F) < 0$ for all $F > 0$: without a positive price spillover, both endogenous-fragility terms push the coordinated foreign-investor position below the competitive position under the ranking conditions of Proposition 4.

Corollary 1 (A wedge can remove appreciation-feedback multiplicity)**.** *Fix g. If τ is such that $\theta B(\tau) \leq \theta\kappa + \lambda(g)S(g,\tau)$, then $H'(F) > 0$ for all $F > 0$. The model therefore*

has a unique positive equilibrium when $A > 0$ and only $F = 0$ when $A \leq 0$; the high-inflow equilibrium cannot coexist with the no-inflow equilibrium.

Proof. See Appendix A. □

2.8 Risk price, tail risk, and amplification

A risk-price analogue. Summarize incentives at the aggregate position by a risk price (mean per unit of risk),

$$\rho(F, g, \tau) \equiv \frac{\mu_X(F, g, \tau)}{\sigma_X(F, g, \tau)} = \frac{A + BF - CF^2}{\sqrt{S + DF^2}}. \tag{31}$$

The numerator is carry net of endogenous crash losses; the denominator is total risk (normal volatility plus jump risk that rises with inflows).

Lemma 3 (Endogenous tail risk lowers the risk price at high inflows)**.** *If $C > 0$ and $D > 0$, then $\rho(F, g, \tau) \to -\infty$ as $F \to \infty$, and there exists $\bar{F}_\rho > 0$ such that ρ is strictly decreasing for all $F \geq \bar{F}_\rho$.*

Proof. See Appendix A. □

The expected return declines quadratically in F (through ηF^2) while volatility grows only linearly, so the crash drag eventually dominates and the per-unit reward to carry collapses in crowded markets.

Tail-risk elasticity. Define the tail event as the crash state and the tail probability $\mathcal{T}(F, g) \equiv \pi(F, g) = \bar{\pi}(g) + \eta F^2$.

Proposition 6 (Inflow elasticity of crash probability)**.** *The log–log elasticity of tail risk with respect to aggregate inflows is*

$$\varepsilon_{\mathcal{T},F}(F, g) \equiv \frac{\partial \log \mathcal{T}}{\partial \log F} = \frac{2\eta F^2}{\bar{\pi}(g) + \eta F^2}. \tag{32}$$

If $\eta F^2 \gg \bar{\pi}(g)$, then $\varepsilon_{\mathcal{T},F} \to 2$: a 1% increase in aggregate inflows raises crash probability by about 2%.

Proof. See Appendix A. □

This convex sensitivity is the heart of the policy concern: once the economy is far enough into the high-inflow region, incremental inflows mainly buy tail risk even while average returns still look attractive.

Amplification, internalized vs. not. The competitive cubic (18) embeds two opposing aggregate feedbacks: carry amplification (inflows raise appreciation, term θBF) and fragility (inflows raise crash loss and variance, terms θCF^2 and λDF^3). The coordination cubic (25) embeds the same feedbacks but with the coefficients $2\theta B$, $3\theta C$, and $2\lambda D$, reflecting full internalization. The contrast is instructive: every place the coordination condition carries a larger coefficient is a place where the private market under-weights the marginal effect of its own crowding. The wedge $\delta(F)$ of Proposition 3 is exactly the sum of these coefficient gaps multiplied by the relevant powers of F.

2.9 Relation to systemic-externality models

The endogenous-fragility channel is related to models in which atomistic choices change an aggregate state that affects other agents. In Lorenzoni [2008], Bianchi [2011], and Mendoza [2010], private borrowing can alter equilibrium prices or collateral capacity in ways individual borrowers do not internalize, motivating macroprudential interventions [Jeanne and Korinek, 2010, Korinek, 2011, Bianchi and Mendoza, 2018]. The analogy here is limited but useful: aggregate foreign positioning raises a common crash probability and a common jump-risk variance.

The exchange-rate appreciation term should be treated differently. In the foreign-investor coordination objective it is a positive return spillover within that sector, but without domestic agents or a collateral/constraint channel it is not a standard welfare-relevant pecuniary externality. The model therefore does not claim to reproduce the host-country welfare results of the inefficient-credit-cycle literature. Its robust normative content is narrower: foreign investors fail to internalize the common fragility their aggregate positions create. A richer general-equilibrium model could add domestic borrowers, exporters, taxpayers, or asset issuers and determine how exchange-rate movements redistribute resources and interact with domestic distortions.

2.10 Comparative statics: the prudential wedge

The wedge τ lowers the foreign investor's expected after-cost local return, but it also scales down the variance of the taxed local payoff. The sign of $\partial F^{\mathrm{CE}}/\partial\tau$ is therefore

not automatic. Define the equilibrium risk-compensation term

$$\mathcal{P}(F, g, \tau) \equiv \frac{\lambda(g)}{\theta} F \sigma_X^2(F, g, \tau). \tag{33}$$

Assumption 3 (Return effect dominates mechanical risk scaling)**.** At the equilibrium branch under consideration,

$$\mathcal{P}(F^{\mathrm{CE}}, g, \tau) < R^*(g) + \kappa F^{\mathrm{CE}}. \tag{34}$$

Proposition 7 (When a higher wedge lowers inflows and tail risk)**.** *At an interior equilibrium with* $H_F(F^{\mathrm{CE}}; g, \tau) > 0$*, if Assumption 3 holds, then*

$$\frac{\partial F^{\mathrm{CE}}}{\partial \tau} < 0, \qquad \frac{\partial \pi(F^{\mathrm{CE}}, g)}{\partial \tau} < 0. \tag{35}$$

Proof. See Appendix A. □

The restriction in Equation (34) is exactly the condition needed to sign the derivative. Economically, the tax-induced reduction in expected return must dominate the fact that multiplying the local payoff by $(1 - \tau)$ also mechanically reduces its variance. When the restriction fails, the model does not claim that a higher wedge necessarily lowers inflows. This qualification is important for both Proposition 5 and the policy discussion.

2.11 Comparative statics: exogenous global funding and risk conditions

The state g is a reduced-form exogenous shifter, not a model-generated global factor. Its role is to ask how a one-country carry market responds when common funding and risk conditions change.

Assumption 4 (Global-condition monotonicities)**.** For higher g (easier global conditions),

$$\frac{dR^*(g)}{dg} < 0, \qquad \lambda'(g) < 0, \qquad \bar{\pi}'(g) < 0, \qquad \sigma_\varepsilon'(g) \leq 0, \qquad \mu_e'(g) \geq 0.$$

These restrictions summarize familiar external forces: easier funding, greater intermediary risk-bearing capacity, and lower global risk are associated with stronger cross-border positions [Adrian and Shin, 2010, Bruno and Shin, 2015, Rey, 2015, Miranda-

Agrippino and Rey, 2020]. They do not imply that the present model explains the cross-country comovement emphasized in the global-financial-cycle literature.

Proposition 8 (Easier global conditions raise inflows)**.** *At an interior competitive equilibrium with* $H_F(F^{\mathrm{CE}}; g, \tau) > 0$*, Assumption 4 implies* $\partial F^{\mathrm{CE}}/\partial g > 0$ *whenever the induced changes raise the private marginal value of the position; in particular this holds under the stated monotonicities because* $A_g > 0$*,* $\lambda'(g) < 0$*, and* $S_g < 0$*.*

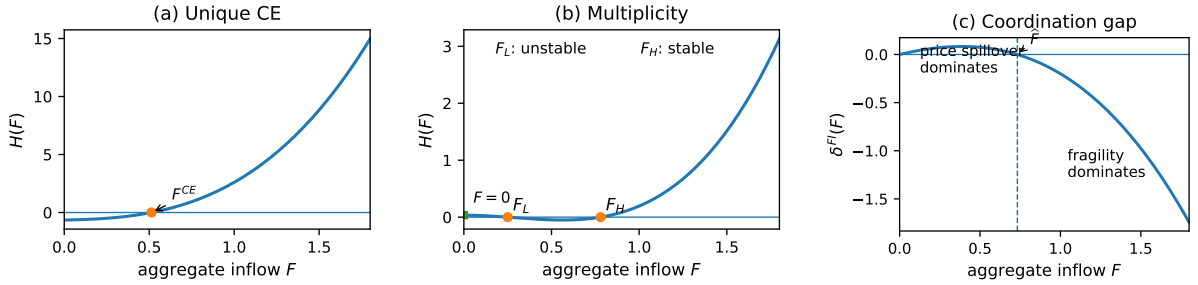
Proof. See Appendix A. □

Thus the model uses the global state as an external source of time variation in funding and risk-taking conditions. Easier conditions expand the position; because $\pi(F, g) = \bar{\pi}(g) + \eta F^2$, that expansion raises the endogenous component of crash risk even if the baseline component $\bar{\pi}(g)$ is lower in good global times.

2.12 Numerical illustration

Figure 1 summarizes the geometry. Panel (a) shows the unique-equilibrium case with positive baseline carry. Panel (b) shows the selective multiplicity case: $A < 0$, the appreciation feedback is strong enough to make $a_1 < 0$, and the local minimum of H lies below zero. The boundary equilibrium and the high-inflow equilibrium are stable under continuous best-response adjustment, while the intermediate root is the tipping point. Panel (c) plots the foreign-investor coordination gap $\delta^{\mathrm{FI}}(F)$. A positive gap means the coordinator's marginal value exceeds the private marginal value; a negative gap means endogenous fragility dominates.

Figure 1: Competitive-equilibrium geometry and the foreign-investor coordination gap



Note: Parameters are illustrative. Multiplicity in panel (b) is generated by the positive appreciation feedback; the convex fragility terms bend $H(F)$ upward and limit the boom. Panel (c) is a foreign-investor-sector benchmark.

The figure also clarifies the role of the policy wedge. Under Assumption 3, a higher τ moves a stable competitive equilibrium toward lower F. In a multiplicity configura-

tion, sufficiently dampening $B(\tau)$ can make $a_1 \geq 0$, eliminating the positive-feedback pair of interior roots. This stabilization result does not rely on interpreting the foreign-investor coordination objective as domestic welfare.

2.13 Beyond exchange rates

Remark 4 (General crowded trades)**.** Nothing in the analysis is specific to currencies. Reinterpret e_1/e_0 as the gross return on any risky asset whose crash probability rises with aggregate positioning—crowded credit, equity-momentum, or volatility-selling strategies—and let χ capture any price-pressure feedback by which aggregate buying supports the asset in normal times. Equations (18)–(26) are unchanged, as are the existence, uniqueness, coordination-gap, and policy results. The framework is thus a general baseline for crowded-trade fragility and within-sector coordination, with the FX application as a leading case.

3 Conclusion

This paper develops a two-date model of crowded currency carry in which atomistic foreign investors take aggregate positioning as given even though aggregate positioning changes the exchange-rate distribution. The private problem is strictly concave; the symmetric competitive equilibrium is a cubic fixed point. That structure delivers a closed-form distinction between three forces: baseline carry, a holding-period appreciation feedback, and endogenous crash fragility.

The first conclusion is about scope. Endogenous crash risk does not, by itself, generate multiple equilibria. With positive baseline carry the competitive equilibrium is unique. A no-inflow equilibrium can coexist with a high-inflow boom only when baseline carry is unattractive and a positive delayed-appreciation feedback is strong enough to overcome baseline balance-sheet and risk costs, with the cubic's local minimum falling below zero. The crash-risk terms then limit the size of the boom. Allowing $\chi \leq 0$ makes the point especially transparent: when there is no positive holding-period appreciation feedback, the equilibrium is monotone and multiplicity disappears. The sensitivity exercise suggests that the required local feedback can overlap with empirical estimates of EM flow-related exchange-rate pressure for some parameterizations, but the full multiplicity region remains deliberately selective.

The second conclusion concerns coordination, not host-country welfare. A coordinator representing the foreign-investor sector internalizes how aggregate positions change

the return distribution faced by that same sector. The resulting marginal gap contains a positive foreign-sector price spillover when $\chi > 0$ and two negative fragility spillovers from expected crash losses and jump-risk variance. Because the model has no domestic side, the appreciation term is not labeled a social benefit or a standard welfare-relevant pecuniary externality. Under the stated uniqueness or concavity conditions, the gap nevertheless provides a rigorous ranking of decentralized positioning relative to the foreign-investor coordination benchmark. When $\chi \leq 0$, the price spillover vanishes or becomes negative and the fragility forces unambiguously push the coordinated position below the competitive one.

The third conclusion is about instruments and qualifications. A wedge on foreign returns reduces inflows and endogenous crash probability only under the explicit return-dominance condition in Assumption 3; the model does not claim an unconditional sign. Under continuity and crossing conditions, the wedge can implement the foreign-investor coordination benchmark and, by weakening the appreciation feedback, can also remove the multiplicity region. The incidence of the wedge matters: tax revenue accruing to the host country is outside the foreign-sector objective, a lump-sum rebate to foreign investors leaves marginal incentives unchanged, and a position-proportional rebate would offset the instrument.

The global financial cycle is best understood here as an exogenous source of common funding and risk conditions. The model maps those conditions into one country's foreign positioning and endogenous FX tail risk; it does not seek to explain cross-country asset-price comovement. Natural extensions would add a domestic sector, market clearing for the local security, and an explicit utility-based welfare criterion. Those additions would be necessary to evaluate the domestic incidence of appreciation, crash losses, tax revenue, and genuinely optimal host-country macroprudential policy. The analytical baseline established here is narrower: crowded foreign positions can endogenously create tail risk, and positive delayed appreciation can turn that crowding into a coordination-driven boom-and-stop regime.

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A Proofs

Proof of Lemma 1. Let the individual investor's objective function be defined as:

$$U_i(f_i; F, g, \tau) = \theta \left(f_i \, \mu_X(F, g, \tau) - \frac{\kappa}{2} f_i^2 \right) - \frac{\lambda(g)}{2} \, f_i^2 \, \sigma_X^2(F, g, \tau)$$

To find the optimal individual position f_i, we first determine the concavity of the objective function with respect to the choice variable. We compute the first and second partial derivatives of U_i with respect to f_i:

The first-order derivative is:

$$\frac{\partial U_i}{\partial f_i} = \theta \, \mu_X(F, g, \tau) - \theta \kappa f_i - \lambda(g) \, \sigma_X^2(F, g, \tau) f_i$$

The second-order derivative is:

$$\frac{\partial^2 U_i}{\partial f_i^2} = -\theta \kappa - \lambda(g) \, \sigma_X^2(F, g, \tau)$$

By assumption, the taste shifter $\theta > 0$, the balance-sheet cost parameter $\kappa > 0$, and the risk aversion coefficient $\lambda(g) > 0$. Furthermore, the variance of the excess return is strictly positive, $\sigma_X^2(F, g, \tau) > 0$. Therefore, it immediately follows that:

$$\frac{\partial^2 U_i}{\partial f_i^2} < 0$$

Because the second derivative is strictly negative globally, the objective function U_i is strictly concave in f_i for any fixed aggregate state (F, g, τ). This guarantees that any local extremum is the unique global maximum.

Next, we solve for the unconstrained optimal position, denoted as f_i^*, by setting the first-order condition to zero:

$$\theta \, \mu_X(F, g, \tau) - f_i^* \left(\theta \kappa + \lambda(g) \, \sigma_X^2(F, g, \tau) \right) = 0$$

Rearranging to isolate f_i^* yields:

$$f_i^* = \frac{\theta \, \mu_X(F, g, \tau)}{\theta \kappa + \lambda(g) \, \sigma_X^2(F, g, \tau)}$$

Finally, we must account for the non-negativity constraint $f_i \geq 0$. Because the objective function is a strictly concave parabola opening downwards, the unconstrained maximum f_i^* is the absolute peak of the function.

- If $f_i^* \geq 0$, the constraint is non-binding, and the optimal constrained choice is $f_i = f_i^*$.
- If $f_i^* < 0$, the unconstrained peak lies in the negative domain. Due to the strict concavity of U_i, the function is strictly decreasing for all $f_i > f_i^*$. Consequently, over the feasible domain $[0, \infty)$, the function achieves its maximum at the boundary $f_i = 0$.

Combining these two cases allows us to express the unique best response compactly using the maximum operator:

$$\mathcal{B}(F; g, \tau) = \max \left\{ 0, \frac{\theta \, \mu_X(F, g, \tau)}{\theta\kappa + \lambda(g) \, \sigma_X^2(F, g, \tau)} \right\}$$

This concludes the proof. □

Proof of Lemma 2. Recall from Section 2.4 that an interior symmetric competitive equilibrium $F^{\mathrm{CE}} > 0$ must satisfy the fixed-point condition derived from the best response function in Lemma 1:

$$F^{\mathrm{CE}} = \frac{\theta \, \mu_X(F^{\mathrm{CE}}, g, \tau)}{\theta\kappa + \lambda(g) \, \sigma_X^2(F^{\mathrm{CE}}, g, \tau)}$$

Rearranging this equation yields the polynomial condition $H(F^{\mathrm{CE}}; g, \tau) = 0$, where the cubic function is defined as:

$$H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0$$

First, we evaluate the polynomial at the boundary $F = 0$. Using the coefficients defined in Equation (19):

$$H(0; g, \tau) = a_0 = -\theta A(g, \tau)$$

By assumption, the taste shifter $\theta > 0$ and the baseline expected excess carry $A(g, \tau) > 0$. Therefore, it immediately follows that $H(0; g, \tau) < 0$.

Next, we examine the asymptotic behavior of $H(F; g, \tau)$ as aggregate inflows F grow

arbitrarily large. The leading coefficient of the cubic polynomial is:

$$a_3 = \lambda(g)D(\tau)$$

Since the risk aversion parameter $\lambda(g) > 0$ and the fragility variance scalar $D(\tau) = (1-\tau)^2 R^2 J^2 \eta > 0$, we have $a_3 > 0$. Because the highest-degree term dominates as $F \to \infty$, it follows that:

$$\lim_{F \to \infty} H(F; g, \tau) = +\infty$$

Because $H(F; g, \tau)$ is a polynomial, it is continuous everywhere on the domain $[0, \infty)$. Since $H(0; g, \tau) < 0$ and $H(F; g, \tau) > 0$ for sufficiently large F, the Intermediate Value Theorem guarantees the existence of at least one root $F^* \in (0, \infty)$ such that $H(F^*; g, \tau) = 0$.

Finally, we must verify that this root constitutes a valid competitive equilibrium according to Definition 1. By construction, $H(F^*) = 0$ implies:

$$F^* \left(\theta\kappa + \lambda(g)\sigma_X^2(F^*, g, \tau) \right) = \theta \mu_X(F^*, g, \tau)$$

Because $\theta, \kappa, \lambda > 0$ and $\sigma_X^2 > 0$, the term in the parentheses is strictly positive. Since $F^* > 0$, it must be true that the right-hand side $\theta \mu_X(F^*, g, \tau) > 0$. Consequently, the unconstrained best response evaluated at F^* is strictly positive, meaning the non-negativity constraint in Lemma 1 does not bind:

$$\mathcal{B}(F^*; g, \tau) = \max\{0, F^*\} = F^*$$

Thus, F^* is a fixed point of the best response map. This confirms the existence of at least one interior symmetric competitive equilibrium $F^{\text{CE}} \in (0, \infty)$. □

Proof of Proposition 1. Recall the competitive equilibrium is characterized by the roots of the cubic polynomial $H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$. From Equation (19), the coefficients are defined as:

$$\begin{aligned} a_3 &= \lambda(g)D(\tau) \\ a_2 &= \theta C(g, \tau) \\ a_1 &= \theta\kappa + \lambda(g)S(g, \tau) - \theta B(\tau) \\ a_0 &= -\theta A(g, \tau) \end{aligned}$$

By the model's baseline parameter restrictions, the risk aversion $\lambda(g) > 0$, the variance scalar $D(\tau) > 0$, the taste shifter $\theta > 0$, and the marginal crash loss $C(g, \tau) > 0$.

Consequently, the two highest-degree coefficients are strictly positive: $a_3 > 0$ and $a_2 > 0$.

We apply Descartes' Rule of Signs to determine the number of positive real roots. We evaluate the sequence of coefficients (a_3, a_2, a_1, a_0) in descending order of degree.

First, consider the case where baseline carry is strictly positive, $A(g, \tau) > 0$. This implies the constant term $a_0 = -\theta A(g, \tau) < 0$. The sign of the linear coefficient a_1 is ambiguous, but we can evaluate both possibilities:

- If $a_1 \geq 0$, the sequence of signs is $(+, +, +, -)$. There is exactly one sign change.
- If $a_1 < 0$, the sequence of signs is $(+, +, -, -)$. There is exactly one sign change.

By Descartes' Rule of Signs, the number of strictly positive real roots is equal to the number of sign changes minus a non-negative even integer. Because there is exactly one sign change in both scenarios, the polynomial $H(F; g, \tau)$ must have exactly one positive real root. Hence, the interior competitive equilibrium is unique.

Second, consider the more general sufficient condition where $a_2 > 0$ and $a_0 \leq 0$. If $a_0 < 0$, the proof of uniqueness holds exactly as above. If $a_0 = 0$ (which occurs when baseline carry $A(g, \tau) = 0$), the polynomial factors as:

$$H(F; g, \tau) = F(a_3 F^2 + a_2 F + a_1) = 0$$

One root is trivially $F = 0$. The remaining positive roots must satisfy the quadratic equation $a_3 F^2 + a_2 F + a_1 = 0$. We again apply Descartes' Rule of Signs to the sequence (a_3, a_2, a_1), knowing $a_3 > 0$ and $a_2 > 0$:

- If $a_1 \geq 0$, the signs are $(+, +, +)$. There are zero sign changes, meaning there are no positive real roots.
- If $a_1 < 0$, the signs are $(+, +, -)$. There is exactly one sign change, meaning there is exactly one positive real root.

Thus, when $a_0 = 0$, the system yields either zero or one strictly positive root. Combining this with the $a_0 < 0$ case proves that whenever $a_2 > 0$ and $a_0 \leq 0$ (which holds naturally when $C > 0$ and $A \geq 0$), there is at most one positive interior equilibrium. □

Proof of Proposition 2. To find the roots of the cubic equation $H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$, we first apply the Tschirnhaus transformation to eliminate the

quadratic term. We substitute $F = y - \frac{a_2}{3a_3}$ into the equation. Expanding the terms yields:

$$0 = a_3 \left[y^3 - y^2 \frac{a_2}{a_3} + 3y \left(\frac{a_2}{3a_3} \right)^2 - \left(\frac{a_2}{3a_3} \right)^3 \right] + a_2 \left[y^2 - 2y \frac{a_2}{3a_3} + \left(\frac{a_2}{3a_3} \right)^2 \right] + a_1 \left(y - \frac{a_2}{3a_3} \right) + a_0.$$

The coefficient on y^2 cancels: $a_3(-a_2/a_3) + a_2 = 0$. Dividing the entire equation by the strictly positive leading coefficient a_3 leaves the monic depressed cubic:

$$y^3 + p\,y + q = 0$$

where gathering the constants produces exactly $p = \frac{3a_3a_1 - a_2^2}{3a_3^2}$ and $q = \frac{2a_2^3 - 9a_3a_2a_1 + 27a_3^2a_0}{27a_3^3}$.

The nature of the roots of this depressed cubic is governed by its discriminant, $\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$.

Case 1: $\Delta > 0$ **(One real root).** By Cardano's method, we let $y = u + v$, which implies $y^3 = u^3 + v^3 + 3uv(u + v) = u^3 + v^3 + 3uvy$. Equating this term-by-term to $y^3 + py + q = 0$ yields the system $3uv = -p$ and $u^3 + v^3 = -q$. Cubing the first equation gives $u^3v^3 = -p^3/27$. Thus, u^3 and v^3 are the two roots of the quadratic equation $Z^2 + qZ - p^3/27 = 0$. Applying the quadratic formula:

$$Z = -\frac{q}{2} \pm \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3} = -\frac{q}{2} \pm \sqrt{\Delta}$$

Taking the cube roots isolates u and v, yielding the single real root:

$$y^\star = u + v = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}$$

The corresponding aggregate inflow is $F^{\mathrm{CE}} = y^\star - \frac{a_2}{3a_3}$.

Case 2: $\Delta < 0$ **(Three real roots).** This is the *casus irreducibilis*, where Cardano's formula requires the cube root of a complex number. Since $\Delta < 0$, it is algebraically necessary that $p < 0$. We use François Viète's trigonometric substitution. Recall the identity $4\cos^3(\theta) - 3\cos(\theta) = \cos(3\theta)$. We substitute $y = u\cos(\theta)$ into the depressed cubic:

$$u^3\cos^3(\theta) + pu\cos(\theta) + q = 0 \implies 4\cos^3(\theta) + \frac{4p}{u^2}\cos(\theta) + \frac{4q}{u^3} = 0$$

Matching coefficients with the identity, we set $\frac{4p}{u^2} = -3$, which yields $u = 2\sqrt{-\frac{p}{3}}$. We

then match the constant term:

$$\frac{4q}{u^3} = -\cos(3\theta) \implies \cos(3\theta) = -\frac{4q}{8(-\frac{p}{3})^{3/2}} = \frac{3q}{2p}\sqrt{-\frac{3}{p}}$$

Because $\Delta < 0$, the argument of the arccosine is strictly bounded between -1 and 1. Solving for θ provides three distinct angles reflecting the periodic nature of the cosine function:

$$\theta_k = \frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{-\frac{3}{p}}\right) - \frac{2\pi k}{3}, \quad \text{for } k = 0, 1, 2$$

This yields three distinct real roots $y_k = 2\sqrt{-\frac{p}{3}}\cos(\theta_k)$, which map to $F_k = y_k - \frac{a_2}{3a_3}$.

Case 3: $\Delta = 0$ **(Repeated real roots).** The algebraic solutions transition seamlessly. If $\Delta = 0$ and $p = q = 0$, then $y = 0$ is a root of multiplicity three. Otherwise, it produces a single real root $y = \frac{3q}{p}$ and a double real root $y = -\frac{3q}{2p}$.

Finally, because all baseline parameters a_0, a_1, a_2, a_3 are real, the shift constant $a_2/(3a_3)$ is purely real. Therefore, any real root y maps to a strictly real aggregate inflow F. Since position sizes cannot be negative, the economically meaningful equilibria are restricted to the non-negative real roots $F \geq 0$. □

Proof of Proposition 3. To derive the foreign-investor coordination gap $\delta^{\text{FI}}(F)$, we compute and compare the marginal objective of the foreign-investor coordinator and the private marginal value of an atomistic investor evaluated at a symmetric allocation $f_i = F$ under $\tau = 0$.

Step 1: Private marginal value at $\tau = 0$**.** From Equation (16), an individual investor i chooses $f_i \geq 0$ taking the aggregate position F as given to maximize:

$$U_i(f_i; F, g, 0) = \theta\left(f_i\,\mu_X(F, g, 0) - \frac{\kappa}{2}f_i^2\right) - \frac{\lambda(g)}{2}\,f_i^2\,\sigma_X^2(F, g, 0).$$

Differentiating U_i with respect to the individual choice variable f_i while treating F, $\mu_X(F, g, 0)$, and $\sigma_X^2(F, g, 0)$ as fixed yields the private first-order derivative:

$$\frac{\partial U_i(f_i; F, g, 0)}{\partial f_i} = \theta\,\mu_X(F, g, 0) - \theta\kappa f_i - \lambda(g) f_i\,\sigma_X^2(F, g, 0).$$

From Equation (15) evaluated at $\tau = 0$, the conditional mean and variance of the per-unit excess return are $\mu_X(F, g, 0) = A_0 + B_0 F - C_0 F^2$ and $\sigma_X^2(F, g, 0) = S_0 + D_0 F^2$.

Evaluating the private marginal payoff at the symmetric configuration $f_i = F$ gives:

$$\begin{aligned}\mathrm{MV}_0^{\mathrm{priv}}(F) &\equiv \left.\frac{\partial U_i(f_i;F,g,0)}{\partial f_i}\right|_{f_i=F} \\ &= \theta(A_0 + B_0F - C_0F^2) - \theta\kappa F - \lambda(g)F(S_0 + D_0F^2) \\ &= \theta A_0 + \big[\theta(B_0 - \kappa) - \lambda(g)S_0\big]F - \theta C_0F^2 - \lambda(g)D_0F^3. \end{aligned} \tag{36}$$

Step 2: Coordinator's marginal objective. The foreign-investor coordinator chooses a common position $f_i = F$ for all $i \in [0,1]$ and internalizes the dependence of $\mu_X(F,g,0)$ and $\sigma_X^2(F,g,0)$ on F. Substituting $\mu_X(F,g,0)$ and $\sigma_X^2(F,g,0)$ into Equation (23) yields the quartic objective in Equation (24):

$$\begin{aligned}W^{\mathrm{FI}}(F;g) &= \theta\left[F(A_0 + B_0F - C_0F^2) - \frac{\kappa}{2}F^2\right] - \frac{\lambda(g)}{2}F^2(S_0 + D_0F^2) \\ &= \theta A_0F + \left[\theta\left(B_0 - \frac{\kappa}{2}\right) - \frac{\lambda(g)}{2}S_0\right]F^2 - \theta C_0F^3 - \frac{\lambda(g)}{2}D_0F^4.\end{aligned}$$

Differentiating $W^{\mathrm{FI}}(F;g)$ with respect to F yields the coordinator's marginal objective in Equation (25):

$$\begin{aligned}W_F^{\mathrm{FI}}(F;g) = \frac{\partial W^{\mathrm{FI}}(F;g)}{\partial F} &= \theta A_0 + 2\left[\theta\left(B_0 - \frac{\kappa}{2}\right) - \frac{\lambda(g)}{2}S_0\right]F - 3\theta C_0F^2 - 4\frac{\lambda(g)}{2}D_0F^3 \\ &= \theta\big[A_0 + (2B_0 - \kappa)F - 3C_0F^2\big] - \lambda(g)\big(S_0F + 2D_0F^3\big) \\ &= \theta A_0 + \big[\theta(2B_0 - \kappa) - \lambda(g)S_0\big]F - 3\theta C_0F^2 - 2\lambda(g)D_0F^3.\end{aligned} \tag{37}$$

Step 3: Derivation of the coordination gap. Subtracting Equation (36) from Equation (37) term-by-term across powers of F gives:

$$\begin{aligned}\delta^{\mathrm{FI}}(F) &\equiv W_F^{\mathrm{FI}}(F;g) - \mathrm{MV}_0^{\mathrm{priv}}(F) \\ &= (\theta A_0 - \theta A_0) + \Big(\big[\theta(2B_0 - \kappa) - \lambda(g)S_0\big] - \big[\theta(B_0 - \kappa) - \lambda(g)S_0\big]\Big)F \\ &\quad + \big(-3\theta C_0 - (-\theta C_0)\big)F^2 + \big(-2\lambda(g)D_0 - (-\lambda(g)D_0)\big)F^3 \\ &= \theta B_0F - 2\theta C_0F^2 - \lambda(g)D_0F^3,\end{aligned}$$

which establishes Equation (26).

Step 4: Chain-rule interpretation. Equivalently, since $W^{\mathrm{FI}}(F;g) = U_i(f_i;F,g,0)\big|_{f_i=F}$, the multivariable chain rule implies:

$$W_F^{\mathrm{FI}}(F;g) = \left.\frac{\partial U_i(f_i;F,g,0)}{\partial f_i}\right|_{f_i=F} + \left.\frac{\partial U_i(f_i;F,g,0)}{\partial F}\right|_{f_i=F} = \mathrm{MV}_0^{\mathrm{priv}}(F) + \left.\frac{\partial U_i(f_i;F,g,0)}{\partial F}\right|_{f_i=F}.$$

Hence, the coordination gap is precisely the uninternalized marginal effect of aggregate positioning F on investor utility evaluated at $f_i = F$:

$$\delta^{\mathrm{FI}}(F) = \left.\frac{\partial U_i(f_i; F, g, 0)}{\partial F}\right|_{f_i=F} = \theta F \frac{\partial \mu_X(F, g, 0)}{\partial F} - \frac{\lambda(g)}{2} F^2 \frac{\partial \sigma_X^2(F, g, 0)}{\partial F}.$$

Substituting $\frac{\partial \mu_X(F,g,0)}{\partial F} = B_0 - 2C_0 F$ and $\frac{\partial \sigma_X^2(F,g,0)}{\partial F} = 2D_0 F$ directly yields:

$$\delta^{\mathrm{FI}}(F) = \theta F(B_0 - 2C_0 F) - \frac{\lambda(g)}{2} F^2 (2D_0 F) = \theta B_0 F - 2\theta C_0 F^2 - \lambda(g) D_0 F^3.$$

This concludes the proof. □

Proof of Proposition 4. We proceed in three steps: first, we link the slope of the coordinator's objective at an interior competitive equilibrium to the coordination gap $\delta^{\mathrm{FI}}(F^{\mathrm{CE}})$; second, we characterize the sign of $\delta^{\mathrm{FI}}(F)$ on $(0, \infty)$; and third, we show that under either condition (i) or condition (ii), the coordinator's objective has a unique global maximizer F^{FI} ranked relative to F^{CE} by the sign of $\delta^{\mathrm{FI}}(F^{\mathrm{CE}})$.

Step 1: Evaluating the coordinator's marginal objective at F^{CE}. Suppose the unregulated ($\tau = 0$) symmetric competitive equilibrium $F^{\mathrm{CE}} \equiv F^{\mathrm{CE}}(g, 0) > 0$ is interior. Because the non-negativity constraint does not bind at an interior equilibrium, the private first-order condition requires that the symmetric private marginal value vanish at $F = F^{\mathrm{CE}}$:

$$\mathrm{MV}_0^{\mathrm{priv}}(F^{\mathrm{CE}}) = 0.$$

Evaluating the coordination gap identity from Proposition 3, $W_F^{\mathrm{FI}}(F; g) - \mathrm{MV}_0^{\mathrm{priv}}(F) = \delta^{\mathrm{FI}}(F)$, at $F = F^{\mathrm{CE}}$ immediately yields:

$$W_F^{\mathrm{FI}}(F^{\mathrm{CE}}; g) = \delta^{\mathrm{FI}}(F^{\mathrm{CE}}). \tag{38}$$

Step 2: Sign characterization of $\delta^{\mathrm{FI}}(F)$ and derivation of $\widehat{F}$. For any $F > 0$, we factor F out of Equation (26) to write:

$$\delta^{\mathrm{FI}}(F) = F\, q(F), \qquad \text{where} \quad q(F) \equiv \theta B_0 - 2\theta C_0 F - \lambda(g) D_0 F^2.$$

Since $F > 0$, the sign of $\delta^{\mathrm{FI}}(F)$ is identical to the sign of the quadratic factor $q(F)$. Recall that under the baseline parameter restrictions, $\theta > 0$, $\lambda(g) > 0$, $C_0 = RJ\eta > 0$, and $D_0 = R^2 J^2 \eta > 0$. We consider two cases depending on the sign of $B_0 = R\chi$:

- *Case 1:* $B_0 \leq 0$. For every $F > 0$, we have $\theta B_0 \leq 0$, $-2\theta C_0 F < 0$, and $-\lambda(g) D_0 F^2 < 0$. Consequently, $q(F) < 0$ and $\delta^{\mathrm{FI}}(F) < 0$ for all $F > 0$. In particular, since

$F^{\text{CE}} > 0$, we have $\delta^{\text{FI}}(F^{\text{CE}}) < 0$.

- *Case 2:* $B_0 > 0$. Differentiating $q(F)$ with respect to F yields $q'(F) = -2\theta C_0 - 2\lambda(g)D_0F < 0$ for all $F \geq 0$, so $q(F)$ is strictly decreasing on $[0, \infty)$. Moreover, $q(0) = \theta B_0 > 0$ and $\lim_{F\to\infty} q(F) = -\infty$. By the Intermediate Value Theorem and strict monotonicity, $q(F) = 0$ admits a unique positive root $\widehat{F} \in (0, \infty)$. Applying the quadratic formula to $\lambda(g)D_0F^2 + 2\theta C_0F - \theta B_0 = 0$ gives:

$$F = \frac{-2\theta C_0 \pm \sqrt{4\theta^2 C_0^2 + 4\lambda(g)D_0\theta B_0}}{2\lambda(g)D_0} = \frac{-\theta C_0 \pm \sqrt{\theta^2 C_0^2 + \lambda(g)D_0\theta B_0}}{\lambda(g)D_0}.$$

Because $\lambda(g)D_0\theta B_0 > 0$, we have $\sqrt{\theta^2 C_0^2 + \lambda(g)D_0\theta B_0} > \theta C_0 > 0$, so the unique positive root is:

$$\widehat{F} \equiv \frac{-\theta C_0 + \sqrt{\theta^2 C_0^2 + \lambda(g)D_0\theta B_0}}{\lambda(g)D_0} > 0,$$

which is Equation (27). Since $q(F)$ is strictly decreasing on $[0, \infty)$, it follows that for all $F > 0$:

$$\delta^{\text{FI}}(F) > 0 \iff 0 < F < \widehat{F}, \qquad \text{and} \qquad \delta^{\text{FI}}(F) < 0 \iff F > \widehat{F}. \tag{39}$$

Step 3: Uniqueness of F^{FI} and global ranking under conditions (i) and (ii). From Equation (25), the coordinator's marginal objective is the cubic polynomial:

$$W_F^{\text{FI}}(F; g) = \theta A_0 + \left[\theta(2B_0 - \kappa) - \lambda(g)S_0\right]F - 3\theta C_0 F^2 - 2\lambda(g)D_0F^3.$$

We now establish that under either condition (i) or condition (ii), $W^{\text{FI}}(F; g)$ possesses a unique global maximizer F^{FI} and $W_F^{\text{FI}}(F; g)$ satisfies the strict single-crossing property around F^{FI}:

- *Under condition (i):* $A_0 > 0$. Evaluating $W_F^{\text{FI}}(F; g)$ at $F = 0$ and as $F \to \infty$ gives:

$$W_F^{\text{FI}}(0; g) = \theta A_0 > 0, \qquad \lim_{F\to\infty} W_F^{\text{FI}}(F; g) = -\infty,$$

since the leading cubic coefficient $-2\lambda(g)D_0$ is strictly negative. Differentiating $W_F^{\text{FI}}(F; g)$ twice with respect to F yields:

$$\begin{aligned} W_{FF}^{\text{FI}}(F; g) &= \theta(2B_0 - \kappa) - \lambda(g)S_0 - 6\theta C_0 F - 6\lambda(g)D_0F^2, \\ W_{FFF}^{\text{FI}}(F; g) &= -6\theta C_0 - 12\lambda(g)D_0F. \end{aligned}$$

Because $\theta, C_0, \lambda(g), D_0 > 0$, we have $W_{FFF}^{\text{FI}}(F; g) < 0$ for all $F \geq 0$, meaning that $W_F^{\text{FI}}(F; g)$ is strictly concave on $[0, \infty)$. A continuous, strictly concave function on

$[0, \infty)$ that is strictly positive at $F = 0$ and tends to $-\infty$ as $F \to \infty$ crosses zero exactly once on $(0, \infty)$: if there existed two distinct positive roots $0 < r_1 < r_2$ with $W_F^{\mathrm{FI}}(r_1; g) = W_F^{\mathrm{FI}}(r_2; g) = 0$, strict concavity of W_F^{FI} on $[0, r_2]$ would imply

$$W_F^{\mathrm{FI}}(r_1; g) > \frac{r_2 - r_1}{r_2} W_F^{\mathrm{FI}}(0; g) + \frac{r_1}{r_2} W_F^{\mathrm{FI}}(r_2; g) = \frac{r_2 - r_1}{r_2} \theta A_0 > 0,$$

a contradiction. Denote this unique positive root by $F^{\mathrm{FI}} > 0$. Then:

$$W_F^{\mathrm{FI}}(F; g) > 0 \quad \text{for } 0 \leq F < F^{\mathrm{FI}}, \qquad W_F^{\mathrm{FI}}(F; g) < 0 \quad \text{for } F > F^{\mathrm{FI}}. \tag{40}$$

Hence $W^{\mathrm{FI}}(F; g)$ is strictly increasing on $[0, F^{\mathrm{FI}}]$ and strictly decreasing on $[F^{\mathrm{FI}}, \infty)$, confirming that F^{FI} is the unique global maximizer.

- *Under condition (ii): W^{FI} is strictly concave on the feasible set.* Strict concavity of $W^{\mathrm{FI}}(F; g)$ implies that $W_F^{\mathrm{FI}}(F; g)$ is strictly decreasing in F and that $W^{\mathrm{FI}}(F; g)$ has a unique maximizer F^{FI} on the feasible set. If $W_F^{\mathrm{FI}}(0; g) > 0$, strict monotonicity of W_F^{FI} implies that $F^{\mathrm{FI}} > 0$ is the unique interior root of $W_F^{\mathrm{FI}}(F^{\mathrm{FI}}; g) = 0$ and satisfies (40). If instead $W_F^{\mathrm{FI}}(0; g) \leq 0$, strict monotonicity implies $W_F^{\mathrm{FI}}(F; g) < 0$ for all $F > 0$, so the unique maximizer is at the boundary $F^{\mathrm{FI}} = 0$; in this case, at any interior competitive equilibrium $F^{\mathrm{CE}} > 0$, we necessarily have $W_F^{\mathrm{FI}}(F^{\mathrm{CE}}; g) = \delta^{\mathrm{FI}}(F^{\mathrm{CE}}) < 0$ and $F^{\mathrm{FI}} = 0 < F^{\mathrm{CE}}$.

Combining Equations (38), (39), and (40) completes the ranking:

1. If $B_0 > 0$ and $F^{\mathrm{CE}} > \widehat{F}$, then $W_F^{\mathrm{FI}}(F^{\mathrm{CE}}; g) = \delta^{\mathrm{FI}}(F^{\mathrm{CE}}) < 0$, which implies $F^{\mathrm{FI}} < F^{\mathrm{CE}}$.
2. If $B_0 > 0$ and $F^{\mathrm{CE}} < \widehat{F}$, then $W_F^{\mathrm{FI}}(F^{\mathrm{CE}}; g) = \delta^{\mathrm{FI}}(F^{\mathrm{CE}}) > 0$, which implies $F^{\mathrm{FI}} > F^{\mathrm{CE}}$.
3. If $B_0 \leq 0$, then $W_F^{\mathrm{FI}}(F^{\mathrm{CE}}; g) = \delta^{\mathrm{FI}}(F^{\mathrm{CE}}) < 0$ for every $F^{\mathrm{CE}} > 0$, which implies $F^{\mathrm{FI}} < F^{\mathrm{CE}}$.

This concludes the proof. □

Proof of Proposition 5. By hypothesis, $F^{\mathrm{CE}}(g, 0) > F^{\mathrm{FI}}(g)$ and there exists $\bar{\tau}$ such that $F^{\mathrm{CE}}(g, \bar{\tau}) < F^{\mathrm{FI}}(g)$. Proposition 7 and the maintained branch conditions imply that $F^{\mathrm{CE}}(g, \tau)$ is continuous and strictly decreasing on $[0, \bar{\tau}]$. The Intermediate Value Theorem therefore gives a $\tau^\star \in (0, \bar{\tau})$ satisfying $F^{\mathrm{CE}}(g, \tau^\star) = F^{\mathrm{FI}}(g)$. Strict monotonicity makes the implementing wedge unique. No welfare statement about the host country is required for this implementation result. □

Proof of Theorem 1. We establish the theorem in four steps: first, we verify the boundary equilibrium at $F = 0$; second, we characterize the unique positive turning point F_m and prove the existence of exactly two interior equilibria $0 < F_L < F_m < F_H$; third, we determine local stability under the adjustment dynamics $\dot{F} = \mathcal{B}(F; g, \tau) - F$; and fourth, we characterize the saddle-node bifurcation.

Step 1: Existence of the boundary equilibrium $F = 0$. From Equation (15), the conditional expected excess return evaluated at zero aggregate positioning is $\mu_X(0, g, \tau) = A(g, \tau)$. Because $A(g, \tau) < 0$ by hypothesis, while $\theta > 0$, $\kappa > 0$, $\lambda(g) > 0$, and $\sigma_X^2(0, g, \tau) = S(g, \tau) > 0$, the unconstrained best response at $F = 0$ is strictly negative:

$$\frac{\theta\, \mu_X(0, g, \tau)}{\theta\kappa + \lambda(g)\, \sigma_X^2(0, g, \tau)} = \frac{\theta A(g, \tau)}{\theta\kappa + \lambda(g) S(g, \tau)} < 0.$$

By Lemma 1, the non-negativity constraint binds at $F = 0$, yielding $\mathcal{B}(0; g, \tau) = \max\{0, \frac{\theta A(g,\tau)}{\theta\kappa + \lambda(g) S(g,\tau)}\} = 0$. Thus, $F = 0$ is a symmetric competitive equilibrium.

Step 2: Turning point F_m and existence of two interior equilibria. From Equations (18)–(19), any interior equilibrium $F > 0$ must satisfy $H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$, where $a_3 = \lambda(g) D(\tau) > 0$ and $a_2 = \theta C(g, \tau) > 0$. Under the assumption $A(g, \tau) < 0$, the constant term is strictly positive:

$$H(0; g, \tau) = a_0 = -\theta A(g, \tau) > 0.$$

Moreover, since $a_3 > 0$, we have $\lim_{F \to \infty} H(F; g, \tau) = +\infty$. Under the strong-feedback condition (29), $\theta B(\tau) > \theta\kappa + \lambda(g) S(g, \tau)$, the linear coefficient is strictly negative:

$$a_1 = \theta\kappa + \lambda(g) S(g, \tau) - \theta B(\tau) < 0.$$

Differentiating $H(F; g, \tau)$ with respect to F yields:

$$\begin{aligned} H'(F; g, \tau) &= 3a_3 F^2 + 2a_2 F + a_1, \\ H''(F; g, \tau) &= 6a_3 F + 2a_2. \end{aligned}$$

Setting $H'(F; g, \tau) = 0$ and applying the quadratic formula gives the stationary points of H:

$$F = \frac{-2a_2 \pm \sqrt{4a_2^2 - 12 a_3 a_1}}{6a_3} = \frac{-a_2 \pm \sqrt{a_2^2 - 3a_3 a_1}}{3a_3}.$$

Because $a_3 > 0$ and $a_1 < 0$, we have $-3a_3 a_1 > 0$, which implies $\sqrt{a_2^2 - 3a_3 a_1} > a_2 > 0$. Consequently, $H'(F; g, \tau) = 0$ has one strictly negative root and one strictly positive

root $F_m > 0$ given by Equation (28):

$$F_m \equiv \frac{-a_2 + \sqrt{a_2^2 - 3a_3a_1}}{3a_3} > 0.$$

Since $H'(0; g, \tau) = a_1 < 0$ and F_m is the unique positive root of $H'(F; g, \tau) = 0$, it follows that:

$$H'(F; g, \tau) < 0 \quad \text{for } F \in [0, F_m), \qquad \text{and} \qquad H'(F; g, \tau) > 0 \quad \text{for } F \in (F_m, \infty). \tag{41}$$

Furthermore, $H''(F_m; g, \tau) = 6a_3F_m + 2a_2 = 2\sqrt{a_2^2 - 3a_3a_1} > 0$, confirming that F_m is the unique local and global minimum of $H(F; g, \tau)$ on $[0, \infty)$.

By hypothesis, $H(F_m; g, \tau) < 0$. Combining this with the monotonicity properties in (41):

- On $[0, F_m]$, $H(F; g, \tau)$ is continuous and strictly decreasing from $H(0; g, \tau) = a_0 > 0$ to $H(F_m; g, \tau) < 0$. By the Intermediate Value Theorem and strict monotonicity, there exists a unique root $F_L \in (0, F_m)$ such that $H(F_L; g, \tau) = 0$.
- On $[F_m, \infty)$, $H(F; g, \tau)$ is continuous and strictly increasing from $H(F_m; g, \tau) < 0$ to $\lim_{F\to\infty} H(F; g, \tau) = +\infty$. By the Intermediate Value Theorem and strict monotonicity, there exists a unique root $F_H \in (F_m, \infty)$ such that $H(F_H; g, \tau) = 0$.

To verify that both F_L and F_H are valid competitive equilibria under Definition 1, note that for each $F^* \in \{F_L, F_H\}$, $H(F^*; g, \tau) = 0$ is equivalent to:

$$F^*\big(\theta\kappa + \lambda(g)\sigma_X^2(F^*, g, \tau)\big) = \theta\mu_X(F^*, g, \tau).$$

Since $F^* > 0$ and $\theta\kappa + \lambda(g)\sigma_X^2(F^*, g, \tau) > 0$, we have $\mu_X(F^*, g, \tau) > 0$, so the non-negativity constraint in Lemma 1 does not bind at F^* and $\mathcal{B}(F^*; g, \tau) = F^*$. Hence, the model has exactly three non-negative competitive equilibria satisfying $0 < F_L < F_m < F_H$. Moreover, the sign of $H(F; g, \tau)$ on $(0, \infty)$ is given by:

$$H(F; g, \tau) > 0 \quad \text{on } (0, F_L) \cup (F_H, \infty), \qquad H(F; g, \tau) < 0 \quad \text{on } (F_L, F_H). \tag{42}$$

Step 3: Local stability under continuous best-response dynamics. Consider the continuous adjustment process $\dot{F} = \mathcal{B}(F; g, \tau) - F$ on $[0, \infty)$. Define the strictly positive denominator $M(F) \equiv \theta\kappa + \lambda(g)\sigma_X^2(F, g, \tau) > 0$ and the unconstrained best response $f^*(F) \equiv \theta\mu_X(F, g, \tau)/M(F)$, so that $\mathcal{B}(F; g, \tau) = \max\{0, f^*(F)\}$ and

$$H(F; g, \tau) = M(F)F - \theta\mu_X(F, g, \tau) = M(F)\big[F - f^*(F)\big].$$

For any $F > 0$, we examine both regimes of the maximum operator:

- If $f^*(F) \leq 0$, then $\mathcal{B}(F; g, \tau) = 0$, which implies $\dot{F} = -F < 0$, while $H(F; g, \tau) = M(F)[F - f^*(F)] > 0$ because $M(F) > 0$, $F > 0$, and $-f^*(F) \geq 0$.
- If $f^*(F) > 0$, then $\mathcal{B}(F; g, \tau) = f^*(F)$, which implies $H(F; g, \tau) = M(F)[F - \mathcal{B}(F; g, \tau)]$, or equivalently:

$$\dot{F} = \mathcal{B}(F; g, \tau) - F = -\frac{H(F; g, \tau)}{M(F)}.$$

Consequently, for all $F > 0$ (whether the non-negativity constraint binds or is slack), we have:

$$\operatorname{sgn}(\dot{F}) = -\operatorname{sgn}\big(H(F; g, \tau)\big). \tag{43}$$

Combining (42) and (43) yields the phase-line dynamics on $[0, \infty)$:

1. On $(0, F_L)$, $H(F; g, \tau) > 0$ and hence $\dot{F} < 0$, so trajectories in $(0, F_L)$ converge monotonically downward to $F = 0$. (In particular, for $F > 0$ sufficiently close to 0, $\mu_X(F, g, \tau) < 0$ by continuity from $\mu_X(0, g, \tau) = A(g, \tau) < 0$, so $\dot{F} = -F$ with right-hand derivative $-1 < 0$ at $F = 0$.) Thus, $F = 0$ is locally asymptotically stable.
2. On (F_L, F_H), $H(F; g, \tau) < 0$ and hence $\dot{F} > 0$. Because $\dot{F} < 0$ on $(0, F_L)$ and $\dot{F} > 0$ on (F_L, F_H) (with local derivative $\frac{d\dot{F}}{dF}\big|_{F=F_L} = -\frac{H'(F_L; g, \tau)}{M(F_L)} > 0$), F_L is unstable and acts as a tipping point separating the basins of attraction $[0, F_L)$ and (F_L, ∞).
3. On (F_H, ∞), $H(F; g, \tau) > 0$ and hence $\dot{F} < 0$. Because $\dot{F} > 0$ on (F_L, F_H) and $\dot{F} < 0$ on (F_H, ∞) (with local derivative $\frac{d\dot{F}}{dF}\big|_{F=F_H} = -\frac{H'(F_H; g, \tau)}{M(F_H)} < 0$), F_H is locally asymptotically stable.

Step 4: Saddle-node bifurcation and sudden stop. Because $H(F; g, \tau)$ and F_m vary continuously with the model parameters, any parameter shift (such as a deterioration in global conditions g or an increase in τ) that raises the local minimum value from $H(F_m; g, \tau) < 0$ to $H(F_m; g, \tau) = 0$ causes the two interior roots F_L and F_H to coalesce at $F = F_m$. At this threshold, $H(F_m; g, \tau) = 0$, $H'(F_m; g, \tau) = 0$, and $H''(F_m; g, \tau) = 2\sqrt{a_2^2 - 3a_3 a_1} > 0$, which are the defining conditions for a saddle-node (fold) bifurcation. Once the parameter moves beyond this threshold so that $H(F_m; g, \tau) > 0$, we have $H(F; g, \tau) \geq H(F_m; g, \tau) > 0$ for all $F \geq 0$. Both interior equilibria vanish, $\dot{F} < 0$ everywhere on $(0, \infty)$, and an economy initially situated at the high-inflow equilibrium F_H undergoes a discontinuous transition (sudden stop) to the unique remaining equilibrium $F = 0$. This concludes the proof. □

Proof of Corollary 1. Fix the global state $g \in (0,1]$ and suppose the policy wedge $\tau \in [0,1)$ satisfies

$$\theta B(\tau) \leq \theta\kappa + \lambda(g)S(g,\tau). \tag{44}$$

(Note that such a $\tau \in [0,1)$ always exists because $\theta\kappa > 0$ is independent of τ, whereas $B(\tau) = (1-\tau)R\chi \to 0$ and $S(g,\tau) = (1-\tau)^2 S(g,0) \to 0$ as $\tau \to 1^-$; in particular, for any $\chi > 0$, every $\tau \geq \max\{0,\, 1-\kappa/(R\chi)\}$ satisfies (44).)

Step 1: Strict monotonicity of $H(F;g,\tau)$ on $[0,\infty)$. Recall from Equation (19) that the equilibrium cubic polynomial is $H(F;g,\tau) = a_3F^3 + a_2F^2 + a_1F + a_0$, with derivative:

$$H'(F;g,\tau) = 3a_3F^2 + 2a_2F + a_1.$$

Under the baseline parameter restrictions ($\theta > 0$, $\lambda(g) > 0$, $\kappa > 0$, $R > 1$, $J > 0$, $\eta > 0$, and $\tau \in [0,1)$), we have $a_3 = \lambda(g)D(\tau) > 0$ and $a_2 = \theta C(g,\tau) > 0$. Moreover, Condition (44) is equivalent to:

$$a_1 = \theta\kappa + \lambda(g)S(g,\tau) - \theta B(\tau) \geq 0.$$

Consequently, for every $F > 0$, we have $3a_3F^2 > 0$, $2a_2F > 0$, and $a_1 \geq 0$, which implies:

$$H'(F;g,\tau) > 0 \qquad \text{for all } F > 0. \tag{45}$$

By the Mean Value Theorem and the continuity of $H(F;g,\tau)$ on $[0,\infty)$, Inequality (45) implies that $H(F;g,\tau)$ is strictly increasing on the entire non-negative domain $[0,\infty)$.

Step 2: Equilibrium characterization depending on the sign of $A(g,\tau)$. We partition the analysis into two exhaustive cases based on the sign of baseline carry $A(g,\tau)$:

- *Case 1:* $A(g,\tau) > 0$. At the boundary $F = 0$, the conditional expected excess return is $\mu_X(0,g,\tau) = A(g,\tau) > 0$, so by Lemma 1 the best response is

$$\mathcal{B}(0;g,\tau) = \max\left\{0,\, \frac{\theta A(g,\tau)}{\theta\kappa + \lambda(g)S(g,\tau)}\right\} = \frac{\theta A(g,\tau)}{\theta\kappa + \lambda(g)S(g,\tau)} > 0.$$

  Hence $F = 0$ is not an equilibrium. For interior equilibria $F > 0$, note that $H(0;g,\tau) = a_0 = -\theta A(g,\tau) < 0$ and $\lim_{F\to\infty} H(F;g,\tau) = +\infty$. Since $H(F;g,\tau)$ is continuous and strictly increasing on $[0,\infty)$, it crosses zero at a unique positive point $F^{\mathrm{CE}} > 0$. As shown in Lemma 2, $H(F^{\mathrm{CE}};g,\tau) = 0$ implies $\mu_X(F^{\mathrm{CE}},g,\tau) > 0$, so $\mathcal{B}(F^{\mathrm{CE}};g,\tau) = F^{\mathrm{CE}}$. Thus, the model possesses a unique competitive equilibrium on $[0,\infty)$, which is interior ($F^{\mathrm{CE}} > 0$).

- *Case 2:* $A(g,\tau) \leq 0$. At the boundary $F = 0$, we have $\mu_X(0,g,\tau) = A(g,\tau) \leq 0$, so the unconstrained best response is non-positive and the non-negativity con-

straint binds:

$$\mathcal{B}(0;g,\tau) = \max\left\{0, \frac{\theta A(g,\tau)}{\theta\kappa + \lambda(g)S(g,\tau)}\right\} = 0.$$

Thus, $F = 0$ is a competitive equilibrium. Furthermore, $A(g,\tau) \leq 0$ implies $H(0;g,\tau) = a_0 = -\theta A(g,\tau) \geq 0$. Because $H(F;g,\tau)$ is strictly increasing on $[0,\infty)$, we have:

$$H(F;g,\tau) > H(0;g,\tau) \geq 0 \qquad \text{for all } F > 0.$$

Therefore, $H(F;g,\tau) = 0$ has no root in $(0,\infty)$. Since any interior competitive equilibrium $F^* > 0$ requires $\mathcal{B}(F^*;g,\tau) = F^* > 0$ and thus $H(F^*;g,\tau) = 0$, no positive equilibrium exists. Consequently, $F = 0$ is the unique competitive equilibrium on $[0,\infty)$.

In both cases, the competitive equilibrium is globally unique on $[0,\infty)$, so a high-inflow interior equilibrium cannot coexist with the no-inflow equilibrium $F = 0$. This concludes the proof. □

Proof of Lemma 3. The proof proceeds in two steps. First, we establish the limiting behavior of the risk price $\rho(F)$ as aggregate inflows grow arbitrarily large. Second, we prove strict monotonicity for large F by analyzing the sign of its derivative.

Step 1: Asymptotic limit of the risk price. From Equation (31), the aggregate risk price is defined as:

$$\rho(F) = \frac{\mu_X(F)}{\sigma_X(F)} = \frac{A + BF - CF^2}{\sqrt{S + DF^2}}$$

To evaluate the limit as $F \to \infty$, we factor out F from both the numerator and the denominator. Since $F > 0$, we can write $\sqrt{S + DF^2} = F\sqrt{\frac{S}{F^2} + D}$. Thus:

$$\rho(F) = \frac{F\left(\frac{A}{F} + B - CF\right)}{F\sqrt{\frac{S}{F^2} + D}} = \frac{\frac{A}{F} + B - CF}{\sqrt{\frac{S}{F^2} + D}}$$

As $F \to \infty$, the terms A/F and S/F^2 vanish to zero. Because $C > 0$, the numerator diverges to $-\infty$. Because $D > 0$, the denominator converges to the strictly positive constant $\sqrt{D}$. Therefore:

$$\lim_{F\to\infty} \rho(F) = -\infty$$

This confirms that at extremely high aggregate inflows, the endogenous crash losses strictly dominate any appreciation benefits, driving the expected risk-adjusted payoff

deeply negative.

Step 2: Strict monotonicity for high inflows. To determine where the risk price is decreasing, we differentiate $\rho(F)$ with respect to F. Applying the standard quotient rule yields:

$$\rho'(F) = \frac{\mu'_X(F)\sigma_X(F) - \mu_X(F)\sigma'_X(F)}{\sigma_X^2(F)}$$

Recall the derivatives of our components:

$$\mu'_X(F) = B - 2CF$$
$$\sigma'_X(F) = \frac{d}{dF}(S + DF^2)^{1/2} = \frac{DF}{\sqrt{S + DF^2}}$$

Substituting these into the quotient rule:

$$\rho'(F) = \frac{(B - 2CF)\sqrt{S + DF^2} - (A + BF - CF^2)\frac{DF}{\sqrt{S+DF^2}}}{S + DF^2}$$

To simplify, we multiply the numerator and denominator by $\sqrt{S + DF^2}$:

$$\rho'(F) = \frac{(B - 2CF)(S + DF^2) - (A + BF - CF^2)DF}{(S + DF^2)^{3/2}}$$

Because the denominator $(S + DF^2)^{3/2}$ is strictly positive for all F, the sign of $\rho'(F)$ depends entirely on the sign of the numerator, which we define as $\mathcal{N}(F)$. Expanding the terms in $\mathcal{N}(F)$ gives:

$$(B - 2CF)(S + DF^2) = BS + BDF^2 - 2CSF - 2CDF^3$$
$$-(A + BF - CF^2)DF = -ADF - BDF^2 + CDF^3$$

Adding these expansions, the BDF^2 terms cancel:

$$\mathcal{N}(F) = -CDF^3 - (2CS + AD)F + BS$$

This establishes that the numerator $\mathcal{N}(F)$ is a cubic polynomial in F. Because $C > 0$ and $D > 0$, the leading coefficient $-CD$ is strictly negative. Therefore, as $F \to \infty$, the cubic function dictates that $\lim_{F\to\infty} \mathcal{N}(F) = -\infty$.

By the definition of the limit, there must exist some threshold $\bar{F}_\rho > 0$ such that for all $F > \bar{F}_\rho$, the numerator $\mathcal{N}(F) < 0$. Consequently, $\rho'(F) < 0$ for all $F > \bar{F}_\rho$. Thus, the risk price is strictly decreasing in the high-inflow region. □

Proof of Proposition 6. By definition, the log-log elasticity of the tail risk probability $\mathcal{T}$ with respect to aggregate inflows F measures the percentage change in crash risk for a 1% change in aggregate positioning. It is computed as:

$$\varepsilon_{\mathcal{T},F}(F,g) \equiv \frac{\partial \log \mathcal{T}}{\partial \log F} = \frac{\partial \mathcal{T}}{\partial F}\frac{F}{\mathcal{T}}$$

Recall the functional form of the crash probability from Equation (3):

$$\mathcal{T}(F,g) = \pi(F,g) = \bar{\pi}(g) + \eta F^2$$

Differentiating this with respect to F yields the marginal increase in crash probability:

$$\frac{\partial \mathcal{T}}{\partial F} = 2\eta F$$

Substituting this derivative and the definition of $\mathcal{T}$ back into the elasticity formula gives:

$$\varepsilon_{\mathcal{T},F}(F,g) = (2\eta F)\frac{F}{\bar{\pi}(g) + \eta F^2} = \frac{2\eta F^2}{\bar{\pi}(g) + \eta F^2}$$

This establishes the exact formula in Equation (32).

Next, we consider the asymptotic behavior in a crowded market where endogenous fragility overwhelmingly dominates the baseline crash risk, meaning $\eta F^2 \gg \bar{\pi}(g)$. To evaluate the limit formally as aggregate inflows grow large ($F \to \infty$), we divide both the numerator and the denominator by the endogenous risk term ηF^2:

$$\varepsilon_{\mathcal{T},F}(F,g) = \frac{\frac{2\eta F^2}{\eta F^2}}{\frac{\bar{\pi}(g)}{\eta F^2} + \frac{\eta F^2}{\eta F^2}} = \frac{2}{\frac{\bar{\pi}(g)}{\eta F^2} + 1}$$

Because the baseline crash probability $\bar{\pi}(g)$ is a fixed parameter for any given global state g, the term $\frac{\bar{\pi}(g)}{\eta F^2}$ vanishes to zero as $F \to \infty$. Evaluating the limit yields:

$$\lim_{F\to\infty} \varepsilon_{\mathcal{T},F}(F,g) = \frac{2}{0+1} = 2$$

Economically, because fragility is quadratic in inflows, the elasticity converges asymptotically to 2. In high-inflow states, every 1% increase in the aggregate position scale drives a proportional 2% increase in the likelihood of a sudden exchange rate collapse. □

Proof of Proposition 7. We establish the comparative statics of the interior symmetric competitive equilibrium $F^{\mathrm{CE}}(g,\tau) > 0$ and the associated crash probability $\pi(F^{\mathrm{CE}}(g,\tau),g)$ in four steps: first, we formulate the equilibrium condition in terms of the symmetric private net marginal value function $\Psi(F;g,\tau) = -H(F;g,\tau)$; second, we compute the partial derivative Ψ_τ and simplify it at the equilibrium allocation using the fixed-point condition; third, we sign $\partial F^{\mathrm{CE}}/\partial\tau$ via the Implicit Function Theorem under Assumption 3; and fourth, we sign the induced change in tail risk $\partial\pi(F^{\mathrm{CE}},g)/\partial\tau$.

Step 1: Equilibrium characterization and regularity. Define the pre-tax holding-period expected gross local return factor $m(F,g)$ and the return-variance factor $s(F,g)$ as:

$$m(F,g) \equiv 1 + \mu_e(g) + \chi F - J\pi(F,g), \qquad s(F,g) \equiv \sigma_\varepsilon^2(g) + J^2\pi(F,g),$$

which depend on (F,g) but are independent of the policy wedge τ. From Equations (7) and (9), the conditional mean and variance of the foreign-currency excess return are:

$$\mu_X(F,g,\tau) = (1-\tau)R\,m(F,g) - R^*(g), \qquad \sigma_X^2(F,g,\tau) = (1-\tau)^2R^2s(F,g).$$

At any interior symmetric competitive equilibrium $F^{\mathrm{CE}} \equiv F^{\mathrm{CE}}(g,\tau) > 0$, the non-negativity constraint in Lemma 1 is slack, and the equilibrium condition $H(F^{\mathrm{CE}};g,\tau) = 0$ in Equation (18) can be written equivalently in terms of the symmetric private marginal value function $\Psi(F;g,\tau) \equiv -H(F;g,\tau)$:

$$\Psi(F;g,\tau) \equiv \theta\big[(1-\tau)R\,m(F,g) - R^*(g) - \kappa F\big] - \lambda(g)F(1-\tau)^2R^2s(F,g) = 0. \quad (46)$$

Because $\Psi(F;g,\tau) = -H(F;g,\tau)$, differentiating with respect to F yields $\Psi_F(F^{\mathrm{CE}};g,\tau) = -H_F(F^{\mathrm{CE}};g,\tau)$. By the maintained hypothesis that $H_F(F^{\mathrm{CE}};g,\tau) > 0$ at the interior equilibrium under consideration, we have:

$$\Psi_F(F^{\mathrm{CE}};g,\tau) < 0. \quad (47)$$

Step 2: Partial derivative with respect to τ and equilibrium substitution. Since $\Psi(F;g,\tau)$ is a polynomial in F and τ, it is continuously differentiable on $(0,\infty)\times[0,1)$. Differentiating Equation (46) partially with respect to τ while holding F and g fixed yields:

$$\Psi_\tau(F;g,\tau) = -\theta R\,m(F,g) + 2\lambda(g)F(1-\tau)R^2s(F,g). \quad (48)$$

The first term $-\theta R\,m(F,g)$ is the marginal reduction in expected carry from a higher wedge, while the second term $+2\lambda(g)F(1-\tau)R^2s(F,g)$ reflects the mechanical reduction in the variance of after-cost local returns as $(1-\tau)^2$ falls.

To evaluate $\Psi_\tau(F;g,\tau)$ at the equilibrium $F = F^{\mathrm{CE}}$, recall from Equation (46) that $\Psi(F^{\mathrm{CE}};g,\tau) = 0$ implies:

$$\theta\big[(1-\tau)R\,m(F^{\mathrm{CE}},g) - R^*(g) - \kappa F^{\mathrm{CE}}\big] = \lambda(g)F^{\mathrm{CE}}(1-\tau)^2R^2s(F^{\mathrm{CE}},g) = \lambda(g)F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau).$$

Dividing both sides by $\theta > 0$ and rearranging isolates the after-cost expected local return at the equilibrium:

$$(1-\tau)R\,m(F^{\mathrm{CE}},g) = R^*(g) + \kappa F^{\mathrm{CE}} + \frac{\lambda(g)}{\theta}F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau). \tag{49}$$

Because $\tau \in [0,1)$, we have $1-\tau > 0$. We can therefore rewrite the two components of Equation (48) evaluated at $F = F^{\mathrm{CE}}$ as:

$$R\,m(F^{\mathrm{CE}},g) = \frac{1}{1-\tau}\left[R^*(g) + \kappa F^{\mathrm{CE}} + \frac{\lambda(g)}{\theta}F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau)\right],$$

and

$$2\lambda(g)F^{\mathrm{CE}}(1-\tau)R^2s(F^{\mathrm{CE}},g) = \frac{2\lambda(g)F^{\mathrm{CE}}(1-\tau)^2R^2s(F^{\mathrm{CE}},g)}{1-\tau} = \frac{2\lambda(g)F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau)}{1-\tau}.$$

Substituting both expressions into Equation (48) at $F = F^{\mathrm{CE}}$ yields:

$$\begin{aligned}
\Psi_\tau(F^{\mathrm{CE}};g,\tau) &= -\frac{\theta}{1-\tau}\left[R^*(g) + \kappa F^{\mathrm{CE}} + \frac{\lambda(g)}{\theta}F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau)\right] + \frac{2\lambda(g)F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau)}{1-\tau} \\
&= \frac{\theta}{1-\tau}\left[-R^*(g) - \kappa F^{\mathrm{CE}} - \frac{\lambda(g)}{\theta}F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau) + 2\frac{\lambda(g)}{\theta}F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau)\right] \\
&= \frac{\theta}{1-\tau}\left[\frac{\lambda(g)}{\theta}F^{\mathrm{CE}}\sigma_X^2(F^{\mathrm{CE}},g,\tau) - R^*(g) - \kappa F^{\mathrm{CE}}\right] \\
&= \frac{\theta}{1-\tau}\Big[\mathcal{P}(F^{\mathrm{CE}},g,\tau) - \big(R^*(g) + \kappa F^{\mathrm{CE}}\big)\Big],
\end{aligned} \tag{50}$$

where $\mathcal{P}(F,g,\tau) \equiv \frac{\lambda(g)}{\theta}F\sigma_X^2(F,g,\tau)$ is the equilibrium risk-compensation term defined in Equation (33).

Step 3: Signing $\partial F^{\mathrm{CE}}/\partial\tau$ via the Implicit Function Theorem. By Assumption 3 (Equation (34)), $\mathcal{P}(F^{\mathrm{CE}},g,\tau) < R^*(g) + \kappa F^{\mathrm{CE}}$, which implies that the bracketed term in Equation (50) is strictly negative. Since $\theta > 0$ and $1-\tau > 0$ for all $\tau \in [0,1)$, it follows that:

$$\Psi_\tau(F^{\mathrm{CE}};g,\tau) < 0. \tag{51}$$

Applying the Implicit Function Theorem to $\Psi(F^{\mathrm{CE}}(g,\tau);g,\tau) = 0$ around (F^{CE},τ) and

combining Inequalities (47) and (51) gives:

$$\frac{\partial F^{\mathrm{CE}}(g,\tau)}{\partial \tau} = -\frac{\Psi_\tau(F^{\mathrm{CE}};g,\tau)}{\Psi_F(F^{\mathrm{CE}};g,\tau)} < 0,$$

because both $\Psi_\tau(F^{\mathrm{CE}};g,\tau)$ and $\Psi_F(F^{\mathrm{CE}};g,\tau)$ are strictly negative, making their ratio strictly positive and $-\Psi_\tau/\Psi_F$ strictly negative.

Step 4: Comparative statics of the crash probability. From Equation (3), the equilibrium crash probability is $\pi(F^{\mathrm{CE}}(g,\tau),g) = \bar{\pi}(g) + \eta F^{\mathrm{CE}}(g,\tau)^2$. Holding the global state g fixed and differentiating with respect to τ via the chain rule yields:

$$\frac{\partial \pi(F^{\mathrm{CE}},g)}{\partial \tau} = \left.\frac{\partial \pi(F,g)}{\partial F}\right|_{F=F^{\mathrm{CE}}} \frac{\partial F^{\mathrm{CE}}(g,\tau)}{\partial \tau} = 2\eta F^{\mathrm{CE}} \frac{\partial F^{\mathrm{CE}}(g,\tau)}{\partial \tau}.$$

Since $\eta > 0$, $F^{\mathrm{CE}} > 0$ at an interior equilibrium, and $\partial F^{\mathrm{CE}}/\partial \tau < 0$, we conclude that:

$$\frac{\partial \pi(F^{\mathrm{CE}},g)}{\partial \tau} < 0,$$

establishing both inequalities in Equation (35). This concludes the proof. □

Proof of Proposition 8. We evaluate the sensitivity of the competitive equilibrium to the global financial cycle using the Implicit Function Theorem. Let the implicit function $\Psi(F;g)$ represent the marginal private value of inflows from the competitive equilibrium condition:

$$\Psi(F;g) \equiv \theta \mu_X(F,g,\tau) - \theta\kappa F - \lambda(g) F \sigma_X^2(F,g,\tau) = 0$$

By the Implicit Function Theorem, the response of aggregate inflows to an improvement in global conditions g is given by:

$$\frac{\partial F^{\mathrm{CE}}}{\partial g} = -\frac{\Psi_g(F^{\mathrm{CE}};g)}{\Psi_F(F^{\mathrm{CE}};g)}$$

As established in the proof of Proposition 7, the denominator $\Psi_F = -H_F$. At the unique, stable interior competitive equilibrium, the equilibrium cubic crosses zero from below, meaning $H_F(F^{\mathrm{CE}}) > 0$. This implies $\Psi_F < 0$. Consequently, the sign of $\frac{\partial F^{\mathrm{CE}}}{\partial g}$ matches the sign of Ψ_g.

We differentiate $\Psi(F;g)$ with respect to g, holding F constant:

$$\Psi_g = \theta\frac{\partial \mu_X}{\partial g} - \lambda'(g)F\sigma_X^2 - \lambda(g)F\frac{\partial \sigma_X^2}{\partial g}$$

We analyze the sign of this expression term-by-term using the structural definitions in Equation (15) and the monotonicities specified in Assumption 4.

First term (Expected return effect): Recall that $\mu_X(F,g,\tau) = A(g,\tau) + B(\tau)F - C(g,\tau)F^2$. Note that the portfolio-balance coefficient $B(\tau)$ and the marginal crash loss $C(g,\tau) = (1-\tau)RJ\eta$ do not depend on g. Thus, $\frac{\partial \mu_X}{\partial g} = \frac{\partial A}{\partial g}$. Expanding the baseline carry $A(g,\tau)$ yields:

$$\frac{\partial A}{\partial g} = (1-\tau)R\big(\mu_e'(g) - J\bar{\pi}'(g)\big) - (R^*)'(g)$$

By Assumption 4, easier global conditions increase baseline appreciation ($\mu_e' \geq 0$), reduce baseline crash probability ($\bar{\pi}' < 0$), and lower funding costs ($(R^*)' < 0$). Thus, $\frac{\partial A}{\partial g} > 0$, making the first term strictly positive: $\theta\frac{\partial A}{\partial g} > 0$.

Second term (Risk appetite effect): Assumption 4 states that easier global conditions lower the effective risk-aversion coefficient, meaning $\lambda'(g) < 0$. Since aggregate inflows $F > 0$ and total variance $\sigma_X^2 > 0$, the negative sign cancels out, yielding a strictly positive contribution: $-\lambda'(g)F\sigma_X^2 > 0$.

Third term (Risk quantity effect): Recall that $\sigma_X^2(F,g,\tau) = S(g,\tau) + D(\tau)F^2$. Since $D(\tau)$ is independent of g, we have $\frac{\partial \sigma_X^2}{\partial g} = \frac{\partial S}{\partial g}$. Expanding the baseline variance $S(g,\tau)$:

$$\frac{\partial S}{\partial g} = (1-\tau)^2R^2\left(\frac{\partial \sigma_\varepsilon^2(g)}{\partial g} + J^2\bar{\pi}'(g)\right)$$

By Assumption 4, both normal-times volatility and baseline crash probability decline with g, meaning $\frac{\partial \sigma_\varepsilon^2(g)}{\partial g} \leq 0$ and $\bar{\pi}' < 0$. Thus, $\frac{\partial S}{\partial g} < 0$. Because $\lambda(g) > 0$ and $F > 0$, subtracting this term yields a strictly positive result: $-\lambda(g)F\frac{\partial S}{\partial g} > 0$.

Summing these three strictly positive components confirms that the marginal private value of inflows strictly increases with easier global conditions:

$$\Psi_g = \underbrace{\theta\frac{\partial A}{\partial g}}_{>0} + \underbrace{(-\lambda'(g))F\sigma_X^2}_{>0} + \underbrace{\lambda(g)F\left(-\frac{\partial S}{\partial g}\right)}_{>0} > 0$$

Returning to the Implicit Function Theorem:

$$\frac{\partial F^{\mathrm{CE}}}{\partial g} = -\frac{(+)}{(-)} > 0$$

Therefore, under the maintained monotonicities, an improvement in the exogenous global funding/risk state expands aggregate carry-trade inflows. □

B A VaR-Style Margin Constraint

Some macroprudential toolkits operate through margins rather than direct taxes. A Value-at-Risk constraint on the position takes the form

$$\mathrm{VaR}_\alpha(f) \le m\,W, \tag{52}$$

where $\alpha \in (0,1)$ is the tail confidence level, $m \in (0,1)$ is margin tightness set by the regulator (lower m is tighter), and $W > 0$ is investor wealth or intermediation capacity. Under the normal-approximation $\mathrm{VaR}_\alpha(f) \approx z_\alpha\, f\, \sigma_X(F,g,\tau)$ with $z_\alpha > 0$ the Gaussian quantile and the position evaluated at the equilibrium aggregate σ_X, the constraint becomes $f\,\sigma_X \le mW/z_\alpha$. Using $\sigma_X = \sqrt{S + DF^2}$ and imposing symmetry $f = F$,

$$F^2\left(S + DF^2\right) \le \left(\tfrac{mW}{z_\alpha}\right)^2 \implies F \le \bar{F}_{\mathrm{VaR}}(g,\tau,m) \equiv \sqrt{\frac{-S + \sqrt{S^2 + 4D(mW/z_\alpha)^2}}{2D}}. \tag{53}$$

The constrained equilibrium is then $F^{\mathrm{CE}}_{\mathrm{VaR}}(g,\tau,m) = \min\{F^{\mathrm{CE}}(g,\tau),\ \bar{F}_{\mathrm{VaR}}(g,\tau,m)\}$. Since $\partial \bar{F}_{\mathrm{VaR}}/\partial m > 0$, tighter margins (lower m) weakly reduce inflows even when the unconstrained equilibrium would be large. The instrument is most consequential in the high-inflow regime, where the endogenous jump term DF^2 makes σ_X steep in F. Under the ranking conditions of Proposition 4, this is also the region in which the negative fragility terms are most likely to dominate the foreign-investor coordination gap. A margin constraint can therefore move positions toward that benchmark when the competitive position is above it, without implying that the rule is optimal for host-country welfare.